University of Europe for Applied Sciences

Declaration on the use of generative Artificial Intelligence (AI) systems

SOHAIL
Name | Family Name

HASSAM
Vorname | First Name

93373509
Matrikelnummer | Student ID Number

Machine Learning for Customer Churn Prediction in Subscription-Based Digital Businesses: A Comparative Analysis of Predictive Models and Customer Retention Factors
Titel Prüfungsarbeit | Title of the exam

I have used the following artificial intelligence (AI)-based tools in the creation of my work:
1. ChatGPT 4o
2. Grammarly
3. Deep Seek

I further declare that

☒ I have actively informed myself about the performance and limitations of the above-mentioned AI tools,
☒ I have marked the passages taken from the above-mentioned AI tools,
☒ I have checked that the content generated with the help of the above-mentioned AI tools and adopted by me is actually accurate,
☒ I am aware that, as the author of this work, I am responsible for the information and statements made in it.

I have used the AI-based tools mentioned above as shown in the table below.

AI-based support tool	Usage	Parts of the work affected	Remarks
EXAMPLE ChatGPT (free Version 4o), OpenAI	• generation of research questions, one of them used • Using it like an examiner	• chapter 1 (introduction) • all chapters	
EXAMPLE Grammarly (free Version 1.2.99.1479), Grammarly	spelling and grammar correction	all chapters	
DeepSeek	Critical checks on content and evaluation on response of the model	All chapters	
…	…	…	

Hamburg, 17.07.2026.

University of Applied Sciences Europe
Iserlohn · Berlin · Hamburg

EIGENSTÄNDIGKEITSERKLÄRUNG / STATEMENT OF AUTHORSHIP

Sohail	Hassam
Name | Family Name	Vorname | First Name

93373509	Machine Learning for Customer Churn Prediction in Subscription-Based Digital Businesses: A Compa
Matrikelnummer | Student ID Number	Titel der Examsarbeit | Title of Thesis

Ich versichere durch meine Unterschrift, dass ich die hier vorgelegte Arbeit selbstständig verfasst habe. Ich habe mich dazu keiner anderen als der im Anhang verzeichneten Quellen und Hilfsmittel, insbesondere keiner nicht genannten Onlinequellen, bedient. Alles aus den benutzten Quellen wörtlich oder sinngemäß übernommen Teile (gleich ob Textstellen, bildliche Darstellungen usw.) sind als solche einzeln kenntlich gemacht.

Die vorliegende Arbeit ist bislang keiner anderen Prüfungsbeh.rde vorgelegt worden. Sie war weder in gleicher noch in ähnlicher Weise Bestandteil einer Prüfungsleistung im bisherigen Studienverlauf und ist auch noch nicht publiziert. Die als Druckschrift eingereichte Fassung der Arbeit ist in allen Teilen identisch mit der zeitgleich auf einem elektronischen Speichermedium eingereichten Fassung.

With my signature, I confirm to be the sole author of the thesis presented. Where the work of others has been consulted, this is duly acknowledged in the thesis' bibliography. All verbatim or referential use of the sources named in the bibliography has been specifically indicated in the text.

The thesis at hand has not been presented to another examination board. It has not been part of an assignment over my course of studies and has not been published. The paper version of this thesis is identical to the digital version handed in.

Hamburg, 17.07.2026	
Datum, Ort | Date, Place	Unterschrift | Signature

Machine Learning for Customer Churn Prediction in Subscription-Based Digital Businesses: A Comparative Analysis of Predictive Models and Customer Retention Factors

Abstract

One of the biggest problems of digital businesses when it comes to subscription-based companies is customer churn since customer maintenance is the main factor in determining profitability, customer lifetime value, and the overall business sustainability. As the amount of customer data increases and machine learning technologies progress, organizations have developed new possibilities to anticipate customers in their actions and act in a proactive way with the retention strategy. This paper presents a study of the effectiveness of machine learning methods in predicting customer churn with the IBM Telco Customer Churn dataset through the sample of 7,043 customer records and 21 variables. Four trained machine learning models, i.e. Logistic Regression, a Decision Tree, a Random Forest, and the Extreme Gradient Boosting (XGBoost) algorithm were used in this study, following comprehensive data cleaning, exploratory data analysis and feature engineering, as well as balancing (class) procedures. The results showed that XGBoost had the highest classification accuracy as it stood at 77.50% and the best recall performance and ROC-AUC score was recorded at Logistic Regression. The most significant predictors influencing customer churn appeared to be type of contract, subscription term, monthly payment, and support services to customers, using feature importance analysis. The article proves the utility of machine learning-powered predictive analytics in terms of customer retention management and makes practical suggestions to sub-based online companies order to enhance customer loyalty and organizational results.

Table of Contents

CHAPTER 1: INTRODUCTION 8
1.1 Introduction to the Chapter 8
1.2 Background of the Study 8
1.2.1 Digital Transformation and Subscription Economy 8
1.2.2 Growth of Subscription-Based Business Models 9
1.2.3 Importance of Customer Retention 9
1.3 Customer Churn in Subscription Businesses 10
1.3.1 Definition of Customer Churn 10
1.3.2 Types of Customer Churn 10
1.3.3 Impact of Customer Churn on Businesses 11
1.4 Problem Statement 11
1.5 Research Aim 12
1.6 Research Objectives 12
1.7 Research Questions 12
1.8 Significance of the Study 13
1.9 Scope of the Study 14
1.10 Structure of the Dissertation 14
CHAPTER 2: LITERATURE REVIEW 14
2.1 Introduction 14
2.2 Customer Churn: Concepts and Definitions 15
2.2.1 Customer Retention and Customer Attrition 16
2.2.2 Types and Causes of Customer Churn 16
2.3 Subscription-Based Digital Business Models 17
2.3.1 Characteristics of Subscription Models 18
2.3.2 Customer Lifecycle Management 18
2.4 Traditional Churn Prediction Methods 19

2.4.1 Statistical Approaches 19
2.4.2 Limitations of Traditional Methods 20
2.5 Machine Learning Approaches for Churn Prediction 20
2.5.1 Supervised Machine Learning 21
2.5.2 Logistic Regression 21
2.5.3 Decision Tree 21
2.5.4 Random Forest 22
2.5.5 XGBoost 23
2.5.6 Comparative Performance of Machine Learning Models 23
2.6 Feature Engineering in Churn Prediction 23
2.6.1 Feature Selection Techniques 24
2.6.2 Feature Transformation and Encoding 25
2.7 Model Evaluation Metrics 26
2.7.1 Accuracy, Precision, Recall and F1-Score 26
2.7.2 ROC-AUC and Confusion Matrix Analysis 27
2.8 Existing Research Findings 27
2.9 Research Gap Identification 28
2.10 Conceptual Framework 29
CHAPTER 3: METHODOLOGY 30
3.1 Introduction 30
3.2 Research Philosophy and Approach 31
3.2.1 Research Philosophy 31
3.2.2 Research Approach 31
3.3 Research Design 32
3.4 Dataset Description 33
3.4.1 Dataset Source 33
3.4.2 Dataset Characteristics 33

3.4.3 Dataset Variables....33
3.5 Variables and Feature Selection....34
3.6 Data Preprocessing....35
3.6.1 Missing Value Treatment....35
3.6.2 Data Encoding....35
3.6.3 Feature Scaling....35
3.6.4 SMOTE Implementation....36
3.7 Exploratory Data Analysis Methods....36
3.8 Machine Learning Algorithms....37
3.8.1 Logistic Regression....37
3.8.2 Decision Tree....**Error! Bookmark not defined.**
3.8.3 Random Forest....**Error! Bookmark not defined.**
3.8.4 XGBoost....**Error! Bookmark not defined.**
3.9 Model Evaluation Techniques....**Error! Bookmark not defined.**
3.10 Ethical Considerations....**Error! Bookmark not defined.**
CHAPTER 4: RESULTS....40
4.1 Introduction....40
4.2 Dataset Overview....40
4.2.1 Dataset Structure....40
4.2.2 Customer Churn Distribution....43
4.3 Exploratory Data Analysis....43
4.3.1 Customer Churn Distribution Analysis....43
4.3.2 Monthly Charges and Customer Churn Analysis....44
4.3.3 Subscription Tenure and Customer Churn Analysis....45
4.3.4 Contract Type and Customer Churn Analysis....46
4.3.5 Internet Service and Customer Churn Analysis....47
4.3.6 Correlation Analysis....48

4.4 Data Preprocessing Results....49
4.4.1 Data Cleaning and Transformation....49
4.4.2 Class Balancing Using SMOTE....51
4.5 Logistic Regression Analysis....52
4.5.1 Model Development....52
4.5.2 Model Performance Analysis....52
4.6 Decision Tree Analysis....54
4.6.1 Model Development....54
4.6.2 Model Performance Analysis....54
4.7 Random Forest Analysis....56
4.7.1 Model Development....56
4.7.2 Model Performance Analysis....56
4.8 XGBoost Analysis....58
4.8.1 Model Development....58
4.8.2 Model Performance Analysis....58
4.9 Model Performance Comparison....60
4.9.1 Comparative Performance Analysis....60
4.9.2 ROC Curve Comparison....60
4.10 Feature Importance Analysis....61
4.10.1 Feature Ranking Analysis....61
4.10.2 Business Interpretation of Feature Importance....61
CHAPTER 5: CONCLUSION....63
5.1 Introduction....63
5.2 Discussion of Findings....64
5.2.1 Customer Churn Behavior Analysis....64
5.2.2 Machine Learning Performance Analysis....65
5.2.3 Feature Importance Interpretation....66

5.3 Comparison with Previous Studies 67
5.4 Practical Implications 68
5.5 Limitations of the Study 69
5.6 Reflection on the Research Process 70
5.7 Future Research Directions 70
References 72

CHAPTER 1: INTRODUCTION

1.1 Introduction to the Chapter

With the introduction of online technologies and data-based business models has created an overwhelming impact on the business organization structure of modern organizations. Digital business, particularly subscription based business has been on the rise due to their ability to generate recurrent revenues, strengthened customer relationships and increased long term business sustainability. One of the most important aspects of subscription realities however is customer retention, as customer dissatisfaction will certainly affect the profitability and organizational activity (AbdelAziz et al., 2025). This chapter offers the background of the research, identification of the research problem, research objectives and questions and the overall outline of the dissertation.

1.2 Background of the Study

The increased popularity of digital technologies in business practices has brought paradigm shifts in how businesses interact with their customers, how they provide their services and how they generate profits. Digital transformation has facilitated businesses to choose new models of operation which focus on customer experience, access to services and long term customer interactions (AbdelAziz et al., 2025). Of these changes, the business model of subscriptions has proven to be one of the most successful, and fast growing in a number of industries, such as software services, digital entertainment, telecommunications, cloud computing, online education, healthcare, and e-commerce.

1.2.1 Digital Transformation and Subscription Economy

Digital transformation can be defined as adoption of digital technologies across the business processes, products, services and the strategies of an organization in an attempt to enhance efficiency in operations and value creation to the customers. The development of cloud computing, artificial intelligence, and big data analytics, mobile technologies, and internet connectivity has hastened the move to digital ecosystems based on customers, rather than transaction models, that were typical of the traditional way of doing business (Adiputra et al., 2025). The companies are increasingly resorting to online platforms to learn more about the customers, researching the consumer behavior and providing tailor-made services.

The largest impact of the digital transformation has been the subscription economy. Contrary to the tech of the olden days where business was based on ownership, subscription businesses

enable individuals to have unlimited access to products and services by paying regularly. With this model the companies can establish a long-term relationship with the customers and simultaneously establish predictable and sustainable lines of revenue (Agrawal, 2025). The subscription economy has been increasing rapidly because it is flexible, convenient, cost effective and personalized to its customers in addition to providing a business with an opportunity to expand in the long term.

1.2.2 Growth of Subscription-Based Business Models

The subscription-based business models have been rampant in the past decade with the introduction of technology and change in the behavior of the consumers. Software as a service (SaaS), stream media, cloud computing, telecommunication, and organizations that deal with offering digital content are migrating to adopt subscription systems as a method of ensuring sustainable customer service (AL-Najjar et al., 2022).

Among the forces that have led to the rise of popularity of subscription services are the increase in internet connectivity, the advent of digital services, consumer need to derive convenience and a preference to consume service over owning a product. Subscriptions would also be beneficial to businesses since the companies are able to better forecast their revenue more easily, extend their customer lifetimes and enable the companies to plan their plans further (Alam et al., 2026). The sustainability of these business models however greatly relies on the loyalty of these customers and decreased attrition of the customers.

1.2.3 Importance of Customer Retention

Retention of customers is a very important issue that determines the success and sustainability of subscription business. The cost of maintaining old customers is often lower than that of acquiring new customers and brand advocacy and improvements in the profitability and lifetime value of the loyal customers are higher than the acquisition costs (Aljifri, 2024). In view of this, customer engagement, customer loyalty programs and customer experience management program are costing companies a fortune.

However this has been a thorn to the majority of subscription-based companies on customer churn. Being aware of the one that will amicably end his/her subscription will assist organizations in establishing retention strategies in advance and reduce on losses. Machine learning and predictive analytics also have opened new opportunities to companies to leverage pattern of customer behavior to improve customer retention rates by conducting data-driven decision-making processes.

1.3 Customer Churn in Subscription Businesses

The customer churn is now one of the most widely-researched issues in customer relationship management and predictive analytics because the churn is directly related to business performance and sustainability. Since a Customer Relationship based on recurrent revenues is based on loyalty, customer retention is a must in digital subscriber environment (Altairey and Al-Alawi, 2024). As a result, the knowledge about the factors affecting customer churn behavior is a strategic interest of the organizations that have to operate in competitive digital markets.

1.3.1 Definition of Customer Churn

Customer churn is a term used to consider customer relationships with a business that are terminated by terminating subscriptions, services, or business. In the context of digital businesses with subscription, customer churn comes into play when either a customer does not want to renew his or her subscription or they just choose to cancel their service contracts (Asif et al., 2025).

Customer churn is a business performance metric that is reported to be paramount in the business as it directly affects revenue generation, customer lifetime value and eventual profitability of the organization (Ayodeji O.J. Ibitoye et al., 2025). Proper prediction of churning will allow companies to determine who has a high likelihood of unsubscribing and take relevant customer retention measures before it is too late.

1.3.2 Types of Customer Churn

The churn of customers could occur under different conditions and be roughly subdivided into voluntary and involuntary churn. A case of voluntary churn is where the customers were not satisfied, price-conscious and shifted to competing products and services and left. On the contrary, involuntary churn is as a result of external factors like payment default, financial problems or technical problems (Chajia & Nfaoui, 2024).

Customer churn can also be divided under active churn where it is customers formally requesting to be canceled, and also passive churn where the customers over time stop the use of the service without necessarily terminating it. Insights on the different types of customer churn will assist companies to come up with an efficient method of customer retention.

1.3.3 Impact of Customer Churn on Businesses

Significant financial, operational and strategic consequences of customer churn to subscription-based businesses. The customer turnover improves the business recurring revenues, customer acquisition expenses and overall profitability. In addition, a high customer churn-rate would force companies to allocate more resources in promotion and customer acquisition costs so as to compensate the revenue lost.

Organizational reputation, level of customer satisfaction, and positioning is also impacted by customer churn. The continuous loss of customers can signify that there are inherent issues in terms of service quality, pricing policy, customer experience or performance of the company (Chang et al., 2024). A decrease in the customer churn has hence assumed strategic goals amid businesses with sustainable growth and competitiveness in the long run in the market.

1.4 Problem Statement

The rapid evolution of the digital businesses based on the sub-model has created a considerable opportunity to create recurent revenue and customer relations. However, the problem of loyalty to customers during a long period of time is one of the most topical of organizations that are engaged in working in the subscription environment. The explicit impacts on the business profitability, operations efficiency, customer lifetime value, and long term sustainability are customer churn. With the upsurge in competition amongst organizations in the market, firms are in the process of prodding the right methods of identifying which customers have the likelihood of canceling their subscriptions.

The retaining methods based more on the descriptive analysis, historical reporting and managerial judgment tend to be much more staid and conservative (Dang Tran et al., 2023). Such techniques are highly useful to find the information, yet, they cannot always identify the complex relations and behavioural patterns which cause the loss of customers. Also, the growing amount and complexity of data about customers generated by online platforms has brought down the efficacy of traditional analytical strategies.

Machine learning is a customer churn prediction tool that has been experiencing a spurt because of the ability to process large volumes of data, identify latent trends and make reasonable predictive associations. Even though there is an increasing trend in the popularity of machine learning applications, it remains unknown how the different algorithms have a comparative level of performance and which of the factors are most probable to herald customer churn (Dang Tran et al., 2023). Consequently, one should explore how effective machine learning

methods can be used in prediction of customer churn in digital businesses with a subscription scheme and what customer features are likely to impact customer retention the most. These are some of the problems that the study will solve; it will develop and deploy different machine learning models of customer data in real life.

1.5 Research Aim

The overall objective of the study is to analyze how machine learning techniques could be effective in predicting the customer churn in online business where subscription occurs. To determine the predictive power, the study will compare some different machine learning algorithms which include: the Logistic Regression, Decision tree, random forest and the extreme gradient Boosting (XGBoost) algorithms. The research will also determine the customer traits and customer behavior that affect customer churn most of all, and thus offer useful insights into supporting customer retention efforts and business decision-making that is informed by data.

1.6 Research Objectives

The objectives of this study are:

- To explore the condition affecting customer churn in online businesses with subscription.
- To examine demographic, behaviour, service related and financial attributes of customers.
- To create models of machine learning to predict customer churn.
- To compare the predictive accuracy of Logistic regression, Decision tree, random forest as well as XGboost algorithms.
- To determine the strongest predictors of customer churn.
- To make viable recommendations on how to retain more customers and make business decisions.

1.7 Research Questions

This research seeks to address the following questions:

RQ1: Can machine learning algorithms be effectively used to predict churn of customers on digital businesses that offer subscriptions?

RQ2: What do customer demographic, behavioural and financial variables most effectively predict customer churn?

RQ3: Which machine learning algorithm has the best predictive accuracy when used to analyse customer churn?

RQ4: In what ways can machine learning-based customer churn predictors models be used to assist customers retention management and strategic business decision-making?

1.8 Significance of the Study

The proposed study offers theoretical and practical work to the domains of customer relationship management, predictive analytics, and business intelligence. Academically, the study has input in the increasing literature on machine learning when it comes to customer churn prediction (Fakhar Bilal et al., 2022). The research involves the analysis of different supervised machine learning algorithms and provided additional empirical evidence of the appropriateness of the forecasting analytics approach to digital business environments involving the use of subscription-based products.

The analysis also aids in the understanding of the relationships between the customer demographic characteristics, usage patterns, financial and customer retention actions. What is even more, the role of analysing the feature significance can actually add to the existing knowledge of the variables that could influence a customer churn decision.

The outcomes may be useful to business managers, business specialists involved in customer relationship, business marketers and business decision-makers in terms of practicality. Accurate customer identification of the customers that are at risk of churning allows organizations to construct particular retention plans, respond to customer tendencies, optimize marketing costs, and increase the overall profitability over the long term.

The analysis could also be used to develop customer retention measures, which are also data-driven considering the analysis of the customer attributes which have been most effective in the churn behavior (Fakhar Bilal et al., 2022). With business models that are founded on subscription-based businesses becoming one of the rising trends in the global scene, predictive analytics, along with machine learning tools, will also be more applicable in terms of ensuring the maintenance of the competitive edge and, thus, the business survivability.

1.9 Scope of the Study

In this research paper, the predictor of churning of customers in online business organizations will be based on prediction using subscriptions and application of supervised machine learning. The dataset used in the study is the IBM Telco Customer Churn which contains information on customer demographics, behavioral factors, service related factors, and financial factors. The investigation is characterized by the data preprocessing steps, the exploration of the data analysis, feature engineering, developing machine learning models, performance analysis, and analysis of features importance.

The reason is the four machine learning algorithms that are discussed in the paper: Logistic Regression, Decision Tree, Random Forest, and Extreme Gradient Boosting (XGBoost). The work is grounded on data on the customers in structured form and issues of binary classification prediction. This study does not dive into deep learning models, sentiment analysis, real-time prediction systems and reinforcement learning strategies (Gegeleso and Ebiesuwa, 2025). In such a way, the conclusions are applicable only to the selected data, and machine learning algorithms that should be applied to this work.

1.10 Structure of the Dissertation

This dissertation is in the form of a dissertation which contains five chapters. In Chapter One, the research background, research problem, research objectives, research questions, significance of research and scope of research are presented. Chapter Two is a comprehensive literature review that allowed searching the articles about the customer churn prediction, subscription-based business model, and methods of machine learning. In Chapter Three, a few snippets of information on the research methodology, data set characteristics, data preprocessing and machine learning methods utilized in the research. Chapter Four gives the results of the analysis, model evaluation results, and feature importances. Lastly, Chapter Five includes a discussion of the findings and gives practical implications, limitations of the research and recommendations on how to do future research.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

One of the most critical studies on the customer relationship management, business intelligence, and predictive analytics has been regarded as customer churn prediction. The increased emphasis on the subscription-business model has increased the need to make

organizations understand customer behavior, conditions that influence customer retention, and creating viable ways to minimize customer churning (Kathi Chandra Mouli et al., 2024). Developments in machine learning and artificial intelligence have equipped organizational entities with effective resources during customer data analysis and forecasting overall customer behaviors with more precision.

Research on customer churn prediction has a wide range of literature across several disciplines, such as marketing, information systems, data science, operations management, and business analytics. The previous studies have investigated various customer churn causes, which included demographic, behavioural, financial, service quality and customer satisfaction index. A range of predictive approaches (both easy accessibility statistical approaches, and more advanced machine learning approaches) have also been researched.

The chapter has critically analyzed the available literature that is tied to customer churn prediction in digital business that involves subscriptions. The talk begins with the theoretical background of customer churn and the customer retention and then continues as the discussion turns into the analysis of subscription business models and customer lifecycle management (Kathi Chandra Mouli et al., 2024). These standard churn prediction methods and machine learning are then contrasted and specifically Logistic Regression, Decision Tree, Random Forest and Extreme Gradient Boosting (XGBoost) are discussed. Lastly, the chapter compares the relative effectiveness of the various predictive methods and the gaps that need to be filled.

2.2 Customer Churn: Concepts and Definitions

Churning of customers has become one of the major subjects of modern day business studies since its effect on the profitability of an organization, customer lifetime value and sustainability in competitiveness is far-reaching. The fact that more and more customers are moving towards the implementation of business models that rely on subscription has brought the issue of customer attrition behavior and the need to provide effective customer retention strategies to higher levels (Liu, Shengdong, et al., 2022). Customer churn is among the complex behavioural phenomena that rely upon a variety of demographic, psychological, financial, and operational factors.

The development of any organizations that operate on subscription basis highly relies on the maintenance of the long term relationship with the customer because their models of development are not based on the model of single payment but the recurring payments made by customers. This has actually resulted in the importance of customer churn prediction to the

critical strategic component of business management (Liu, Shengdong, et al., 2022). Customer churn prediction also enables the companies to know which are the high-risk customers to use customer retention and put in advance measures to intervene.

Customer churn is not just a straightforward concept of the end of business relations. Customer attrition usually projects bigger sensual organizational problems in relation to customer contentment and quality of services to be rendered, pricing, consumer experiences, rivalry and changing consumer lifestyles. Customer churn therefore, is a complicated zone that entails the customer action and the market as well as the performance of the organization.

2.2.1 Customer Retention and Customer Attrition

There is a process known as Customer retention that the organization will find itself in a position of retaining customers that are already customers of an organization by giving them consistency, quality services and satisfactory experiences with the customers (Moch Yusuf Asyhari et al., 2025). A high rate of customer retention is normally associated with a high rate of profitability, an increase in the customer lifetime value, increase in brand loyalty and low customer acquisition costs.

The opposite of this phenomenon and referred to as customer churn as well as customer attrition is the term used when customers break business relations with an organization. Customer attrition can be as a result of dissatisfaction, changes in the market, financial reasons, market substitutes, or changes in preferences of customers. A loss of customers in subscription businesses direct punches on consistent revenue streams and has negative implications to performance of an organization.

It is constantly found that it is much more cost-effective to keep customers than it is to acquire them (Naga et al., 2025). Consequently, customer retention management through the use of customer relation management systems, loyalty programs, personalized services and predictive analytics technologies is turning out to be the concern of organizations.

2.2.2 Types and Causes of Customer Churn

The customer churn can be categorised into different categories under different circumstances where the customer leaves. There are two broad types of churn, i.e., voluntary and involuntary churn. Voluntary churn is a situation where customers cancel their subscriptions voluntarily as they are dissatisfied or have issues with the pricing of the service, or with the experience, or

with competitive offers. Unintentional churn: this can occur due to an external set of factors such as default of payments, monetary issues, or administrative errors or technical issues.

There can also be characterized churning of customers into active and passive churn. Active churn occurs when the customers make a statement that they are terminating services and passive churn occurs when the customers reduce the use of the services without terminating subscriptions (Prakash, 2024).

There are many causes of customer churn behavior. Finance tends to influence customer choices, such as prices, how they would like to be billed; the perceived value. Service related factors that also contribute greatly to Customer retention include the quality of customer service, technical reliability, service availability and performance of the products (Prakash, 2024). Other behavioural habits such as frequency of use of the service, contact with the customer, time of subscription and shopping habits have also been observed to be important predictors of customer churn.

Understanding the various reasons and types of churn amongst customers enables organizations to develop customer specific retention procedures and also develop customer relationship management initiatives.

2.3 Subscription-Based Digital Business Models

Buffer-based business models, which are often very common in modern online economies, have become more important in that they enable the creation of a consistent flow of revenues, improvement of communication with customers, as well as the opportunity to take business decisions long-term. Subscription businesses, in comparison to the traditional transactional business model whereby the business makes contact with the customer only once, entails the ongoing customer relationships and the constant service provision to the customer (Prakash, 2024).

The rise of digital technologies have improved the application of subscription services in the different sectors like software development, online entertainment, online education, cloud services, telecommunication, health services and e-commerce. The business models give the organization stable revenues and give the customers flexibility in terms of payment plans in order to access goods and services.

The competitive advantages of any subscription companies achieved due to their ability to accumulate customer data on a frequent regularity and utilize customer awareness to increase

service delivery performances, customize experiences, and simplify customer retention strategies are enormous (Ramesh et al., 2024). Customer satisfaction and customer retention management of subscription models are however critical in the long run success of subscription models.

2.3.1 Characteristics of Subscription Models

Business models that are based on subscription have a number of unique features that distinguish them with the traditional business models. The recurrent nature of revenue that is generated by periodic payment by the customers happens to be among the most crucial features. It is a system of revenues, which enables organizations to increase financial predictability and strategic long-term planning approaches.

Subscriptions are also paid with emphasis on relationship with the customers in the long term rather than a single payment. A lifetime value of customers will become a key performance indicator because of the first cut penetration of long-term customer interactions and profitability (Rasool et al., 2025). In addition, the subscription system assists in organizing a harmonized service enhancement, personalized customer experience, and ongoing communication with customers.

Scalability, the availability of customer data, pricing models, among the potential of customization of services are the other features. These features aid organizations in making fast time adjustments with the changing customer preference and market conditions.

2.3.2 Customer Lifecycle Management

Customer intensity management may be construed as the scientific customer relationship management procedure at various phases in customer treatment procedure. Customer life cycle typically covers acquisition and on boarding of customers, customer interactions, customer retention, customer loyalty creation and customer turnover.

Customer lifecycle management assists organisations to obtain the highest lifetime value and lessen customer churn. During the acquisition stage, the firms are concerned with the fact that they should appeal to new clients through promotions and marketing strategies. Second, onboarding processes will work on familiarizing customers with the services and products and improving the initial customer experiences.

One of the most crucial steps is the retention step since the level of customer loyalty and commitment is an important determinant of business performance. Companies are shifting

towards using predictive analytics and machine learning to track customer behavior in the lifecycle, and to predict customer potential to churn (Shaikhsurab & Magadum, 2024). Customer life cycle management therefore is a very important part of subscription based business operations and customer retention management in strategies.

2.4 Traditional Churn Prediction Methods

Prior to the advent of machine learning technologies, predicting customer churn was mostly based on the use of conventional statistical and analytical methods. The methodologies focused at determining the customer attributes against customer attrition outcomes applying statistical modelling software. Although the traditional methods had their advantages in educating on customer behavior, they might be constrained in predicting behavior when manipulating complicated customer data.

Traditional churn prediction continues to play a major role in customer analytics because of the intuitive outcomes that they generate, and the capability to give a roadmap behind a predictive model configuration (Shaikhsurab & Magadum, 2024). The increasingly complex nature of customer behavior, however, and blistering expanding customer data has revealed the limitations of the past treatments of analysis.

2.4.1 Statistical Approaches

The past has seen different statistical approaches that have been applied in predicting churn of customers. Logistic Regression has been one of the most utilized methods due to its simplicity, interpretability and above all, has served well as a method in the binary issues of classification. Logistic Regression estimates the likelihood of customer churn basing on connections among forecastor variables and customer results.

The other traditional approaches include linear regression models, discriminant analysis, survival analysis and Bayesian classification wherein models. These methods permit the researcher to find statistically significant relationships between customer characteristics and churn behavior and are correlated with model outputs which are interpretable.

Critical conventional forecasting technique was also transformed into decision tree analysis because of its ability to produce rule of classification that was easily understandable and represented customer decision-making pathways (Siddiqui et al., 2024). The use of statistical methods in the creation of customer analytics and in aiding predictive research serves as a crucial benchmark or base model.

2.4.2 Limitations of Traditional Methods

The conventional churn prediction techniques, despite their historical significance have a number of drawbacks when used to address modern customer analytics issues. The assumption made between the variables of linear relationships is one of the limitations (Sina Mirabdolbaghi and Amiri, 2022). Customers usually behave in a non-linear manner, where interaction is complex and can not be properly learnt by traditional statistical models.

The traditional methods also show a poor performance with large datasets having high dimensional variables in them, missing values and suffer when there is an interaction of variables. These restrictions are only getting increasingly troublesome as the amount of data that we provide about customers is increasing.

The other noteworthy limitation is the predictive performance. The methods of conventional statistics tend to perform worse than the state-of-the-art machine learning methods in terms of their theories of prediction since they cannot identify meaningful patterns in high amounts of customer information. Moreover, traditional methods are more likely to assume a lot regarding distributions of information, and variable independence besides relating to statistics (Suh, 2023).

As a result, a weakness of traditional churn prediction techniques has been the impetus to researchers and practitioners to consider machine learning techniques, which are more flexible, have better predictive capabilities, and greater analytical power.

2.5 Machine Learning Approaches for Churn Prediction

The machine learning has emerged as one of the most effective way to forecast customer churn since it enables measuring large volumes of data, revealing the complex trends, as well as making the most accurate forecasts (Usman Gani Joy et al., 2024). Compared to the conventional statistical techniques, machine learning algorithms have the ability to learn relationships between variables automatically, and do not need numerous assumptions about the distribution of data or about the interactions between variables.

The availability of more customer information, the development of more computing power, and artificial intelligence have contributed to a rise in the use of machine learning techniques in managing customer relations and predicting analytics. Other machine learning models demonstrated a great level of success to give IDs of the customers that are at risk of churning and combined with customer retention strategies.

2.5.1 Supervised Machine Learning

Supervised machine learning is the most used machine learning technique in the field of customer churn prediction. In supervised learning algorithms, the previously known customer data with known outcome variables are used to construct a prediction model which can be applied to predict customer behaviour in future.

Training machine learning models on labeled datasets, checking the performance of those models and measuring predictability with the aid of multiple performance metrics are the steps involved in the supervised learning process (Usman Gani Joy et al., 2024). The rational option of customer churn prediction is supervised learning because the answers will either be binary (as customers remain subscribed or discontinue their services as a result).

Motivation Supervised machine learning techniques have several benefits, such as high predictive accuracy, flexibility, scalability and the capability to model complex relationships (Xiahou & Harada, 2022). As such, supervised learning algorithm has been in the forefront of the analysis in the present studies on customer churn prediction.

2.5.2 Logistic Regression

The Logistic Regression-based machine learning model is one of the simplest yet holds interpretability and computational efficiency making it one of the most common to be used in churn prediction of customer. Logistic Regression approximates the likelihood of customer churn, by fitting the association among the predictor variables and a discrete outcome variable.

The initial advantage of Logistic Regression is that it can be interpreted. Companies can easily understand the impact of individual customer features on the likelihood of churn, and make improved business choices (Zhang et al., 2022). Moreover, Logistic Regression does well with variables that have relatively linear relationships amongst them.

The Logistic Regression even though has more complex algorithms in comparison to its alternatives in machine learning, nonetheless, remains to play a strong role in the model comparison because it is a robust, straightforward, and viable model (Zhang et al., 2022). In the problems of customer churn prediction, numerous empirical experiments have shown a satisfactory level of predictive power in the usage of the Logistic Regression.

2.5.3 Decision Tree

In Decision Tree algorithms, the customer observation is classified by recursively splitting the datasets based on decision rules which are based on the predictor variables. The concept of the

Decision Trees is particularly attractive as the hierarchy of classification acquired may be extremely contrary, hence, it becomes straightforward to define understand the business and arrive at a decision.

The decision tree models have various strengths such as being easy to implement, it can be visualized, and it is flexible to work with numerical and categorical variables (Zhou et al., 2023). They also need fairly minimal data preprocessing and are able to detect non linear relationships within attributes of the customers.

Nonetheless, there are a number of weaknesses in the Decision Trees. When applied on detailed databases of customers, single trees are compared with experience overfitting and high variance and poor generalization behaviours. Therefore, despite the fact that Decision Trees are still valuable means of analysis, the decision tree can often be less predictive in accuracy compared to ensemble machine learning techniques.

2.5.4 Random Forest

Random Forest is a form of ensemble machine learning algorithm, which is able to overcome numerous shortcomings of individual decision trees. When constructing the algorithm, one takes random subssets of the observations and variables to construct a few decision trees that are in turn combined by voting by using ensemble processes to come up with their prediction. There are some significant benefits of random forest in predicting churn of customers (Zhou et al., 2023). To start with, ensemble structure minimizes overfitting and enhances the performance of model generalization. Second, the relationship and the complex interaction of features among the features that tend to determine customer behavior are well developed under the Rand Forest. Third, the algorithm has valuable feature scores which can be used by businesses to make sense of things.

In numerous journal articles, it was discovered that Random Forest is among the most convenient machine learning models to forecast customer churn. The fact that it easily copes with large datasets, can deal with missing values and creates sound predictions have also worked in its favor in business analytics applications (Usman Gani Joy et al., 2024). Moreover, Random Forest has good performance in various assessment measures which include accuracy, precision, recall and ROC-AUC. Accordingly, the algorithm has become part of the traditional predictive customers analytics and customer retention management.

2.5.5 XGBoost

XGBoost or Extreme Gradient Boosting is one of the most advanced and efficient machine learning algorithms nowadays and can be applied to process structured data. XGBoost is in addition of the old methods of gradient boosting involving advanced optimization method, regularization measures and performance improvements on the computational efficiency.

XGBoost has become very popular due to its outstanding predictive performance, scaled and high ability. The algorithm also constructs the decision trees recursively at the cost of minimizing the errors in predictions by gradient optimization (Usman Gani Joy et al., 2024). Also, mechanisms of regularization minimize overfitting and enhance the generalization of the models.

Using XGBoost has continually shown better results than most traditional and machine learning methods in customer churn prediction research. The algorithm can effectively approximate nonlinear relations, and also missing data, and is highly accurate in predictions. XGBoost possesses a high predictive power and is often used in many industries including telecommunication, banking, insurance, healthcare and online subscriptions services.

2.5.6 Comparative Performance of Machine Learning Models

Experimental studies of machine learning algorithms on various datasets still reveal the predictive accuracy depends on the specifics of the dataset, heuristics of feature engineering, and the evaluation methods. However, ensemble learning, such as Random Forest and XGBoost, is more likely to work better compared to traditional statistical models and the single decision tree models.

Logistic Regression remains interpretable and good because of its bases, and Decision Trees are clear about their decision guidelines (Zhang et al., 2022). Random Forest and XGBoost are however said to have the strongest customer churn predictive accuracies and robustness and business applicability and are thus one of the most recommended to apply in predicting customer churn in digital business context based on subscription services.

2.6 Feature Engineering in Churn Prediction

The feature engineering step is one of the most essential ones in respect of machine learning model development as the predictive performance directly correlates with the quality and relevance of the input variables. In customer churn prediction studies, feature engineering entails the discovery of specific features, selection of specific features and their transformation

and optimization to enhance the accuracy of the model, render it interpretable and enhance its generalisation ability (AbdelAziz et al., 2025). Although the successful machine learning algorithms possess effective predictive capabilities, many studies have demonstrated that the level of closely successful feature engineering processes greatly affects the model performance.

Customer churn datasets tend to be highly dense in demographic and behavioral variables, as well as in transactional and financial variables, as well as service related variables. However, it does not happen that the variables are equal in terms of predictive performance. Researchers and practitioners, based on this, use the feature engineering method to determine the most valuable variables, with less irrelevant complexity and less computational costs.

The feature engineering concept also enables business to locate unknown relationships between the attributes of the customers that are not readily accessible in raw data. The process helps in enhancing business insight and decision making ability as well as the predictive accuracy (AbdelAziz et al., 2025). The element of good feature engineering enables the development of good predictive models that can identify the churn behavior of the customer in a more efficient and time-saving approach.

2.6.1 Feature Selection Techniques

The process of the identification of the variables with the greatest significance as predictors, followed by the elimination of the variables that are redundant, irrelevant or introduce noise is known as feature selection. The main aim of feature selection is to enhance accuracy of the model, decrease computational complexity, decrease overfitting and improve interpretability of the model. There are several feature selection algorithm that are applied in churn predicting of work in regard to customers (Adiputra et al., 2025). Filter methods are techniques that assess variables to statistical measures like correlation coefficients, information gain, mutual information and chi-square statistics. These techniques identify such variables that are highly correlated with churn outcomes of the customers and do not require large number of computational resources.

Wrapper methods actually combine variables through multiple machine learning model trainings, finding the subset of features which yield the most predictive value. In predictive analytics studies, recursive feature elimination is one of the most well used wrapper methods. Wrapper methods can be highly predictive, but the methods require a lot of computing resources. Embedded methods also have a feature selection when training models (Agrawal, 2025). Random Forest and XGBoost are machine learning algorithms that automatically

establish the significant variables by considering value of the features. Their popularity has been due to the capabilities of these methods to trade off predictive performance and computational efficiency.

Type of contract, period of subscription, money paid per month, the mode of payment, communication with the customer support and patterns of usage of these special services in the customer churn prediction studies have always been found to significantly impact the customer retention rate.

2.6.2 Feature Transformation and Encoding

Transformation of raw data into either a format understandable by machine-learning is known as feature transformation. Customers datasets typically also contain varying types of variables, scales, and measurement instruments that have to be changed to a new format that can be converted and are subjected to transformation processes before being useful to the prediction algorithms.

One of the most essential techniques of transformation is the data normalization and standardization. The operations of standardization transform the variables to similar scales that the impact of the variables with high numerical scores would not have a disproportionate influence on the model performance. The feature scoring is the most crucial one when implementing their algorithms such as the Logistic Regression, distance based machine learning algorithms. The other important problem involved in customer churn prediction is categorical variables (Babatunde et al., 2023). Seeing that machine-learning algorithms may often require their feeding with numerical data, categorical data must be converted with the assistance of encoding algorithms.

Other transformation processes comprise of missing value imputation, logarithmic transformation, discretization and feature combination. The techniques increase the quality of data, noise minimization, and predictive quality. According to the latest studies, effective feature transformation is a long road towards improving the work of machine learning models since it enables algorithms to uncover some latent patterns and relationships in customer information (Babatunde et al., 2023). The feature transformation, in its turn, is one of the keys to the churn prediction frameworks of customers.

2.7 Model Evaluation Metrics

Assessment of machine learning models is also another essential step in the prediction of customer churn research as it has a direct influence on the business decision-making and customer retention. The problems of customer churn prediction are binary classification problems and have to be evaluated using multiple metrics of evaluation in order to provide a comprehensive understanding of model reliability and classification accuracy of customer churn prediction.

The different evaluation metrics can provide different clues about the model performance. Using any individual measure of performance can also give out misleading results, especially in the analysis of uneven sets of customers where the retained percentage is much higher compared to the churned customer (Han, 2024). Researchers, in turn, usually employ numerous evaluation measures to be in a position to obtain a comprehensive perspective of the predictive ability.

Basic evaluation metrics of models can also be used to compare machine learning algorithms. Implementation of standardized performance measures enable the researchers to establish the strengths and weaknesses of the predictive models since it determines whether the predictive models are applicable in the business in real-life setting.

2.7.1 Accuracy, Precision, Recall and F1-Score

Accuracy is one of the most commonly-used performance measures in the area of machine learning studies. Accuracy Provides a measure of the distribution of the frequency of correct observations of an overall number of observations. Giving informative information on predictive performance but when a high imbalance in the form of classes is present, accuracy creates misleading estimations (Gegeleso & Ebiesuwa, 2025).

The rate of positive predictions is estimated using precision to tell how many true positive predictions the model makes are compared to the number of positive predictions the model makes. High precision implies that the model of prediction has relatively small falsely positive classification (Han, 2024). Precision in the context of customer churn prediction indicates how reliable the predictions can be about the possibility of a customer ceasing subscription.

Earlier on, we have defined recall also referred to as sensitivity which will determine the percentage of the actual churn customers that the predictive model will accurately identify. The

reason why recall is significant in customer retention management is that not being able to recognize high-risk customers will lead to huge financial losses.

F1-score is an evaluation measure, which simultaneously assesses the precision and recall, the F1-score is constructed by harmonic mean of evaluation measures. It is a calculated predictive performance which is particularly relevant when a false positive, as well as a false negative, is an important factor in the business (Gegeleso & Ebiesuwa, 2025).

2.7.2 ROC-AUC and Confusion Matrix Analysis

One of the most commonly accepted measures of evaluation metrics of binary classification problems is the Receiver Operating Characteristics Area Under the Curve (ROC-AUC). The relationship between the true positive rates and the false positive positions with each of the different levels of classification is presented by the ROC curve (Gegeleso & Ebiesuwa, 2025). The difference between the churn and non-churn customers of the model is the area under the ROC curve.

The range of scoring is 0-1, but the higher the score, the more power of prediction of the ROC-AUC. The values greater than 0.80 are usually indicated as that value which exhibits a high degree of classification.

The confusion matrixs can provided substantial information regarding the classification outcomes, by displaying the prevalence of true positives, true negatives, false positive and false negatives. Through confusion matrix analysis, a researcher is able to identify some of the strengths and weaknesses of the predictive models, and the business can make sense of discrepancies in predictions.

Together, ROC-AUC analysis and confusion matrix evaluation will provide a comprehensive insight into the predictive results and assistance in finding the appropriate machine learning models to apply to the customer churn prediction issue.

2.8 Existing Research Findings

Customer churn prediction is an area of research that has increased immensely over the past two decades due to the entry of competition in customer-related sectors, and new predictive analytics models. The behavior of the customers churn studied in telecommunications, banking, insurance, software services, e-commerce, healthcare, and digital subscription platform was the focus of several studies (Hasan, Siam, et al., 2024). Early research of customer churn merely involved the standard statistical analysis, including the Logistic Regression,

discriminant analysis and survival analysis. These studies formed the basis of the demography of customers, sequences of service use and monetary parameters to ascertain customer retention results. It was particularly the Logistic Regression that gained popularity because of the absence of challenging interpretability and the ability to efficiently compute it.

With the growing sophistication of customer dataset, students started to use machine learning methods that can discover nonlinear connections and latent patterns. The potential to generate the explainable decision structure and advance business understanding helped the popularity of the Decision Tree algorithms. Still, subsequent studies indicated that single decision trees tend to be over-fitted and their level of prediction is usually lower than that of more intricate customer behavior patterns. The emergence of ensemble learning methods was an important step towards customer churn prediction studies (Hasan, Siam, et al., 2024).

Gradient boosting algorithms have in more recent times emerged as some of the best predictive algorithms in terms of customer churn analysis. XGBoost has garnered special interest because of its superior predictive power, computational speed, and resiliency. Studies in a telecommunication, banking, subscription and online trading environment have continuously reported higher performances with the use of XGBoost algorithms (Kathi Chandra Mouli et al., 2024). It has also been mentioned that feature engineering and customer behavior analysis are vital in research results.

Several comparative studies have compared the time of different algorithms of machine learning in churn prediction within a customer context. These studies all suggested that ensemble learning methods are supreme to the conventional traditional statistical methods, assuming that they can be deemed to outperform in terms of predictive accuracy and strength. However, the researchers have also drawn people's attention to the fact that the work of the model also largely depends on the type of datasets, the feature selection process, the preprocessing process and its evaluation (Liu, Shengdong, et al., 2022). Recent studies have made the explainable artificial intelligence and feature importance analysis, and business interpretability more popular an area of focus. It is not just that the organizations require correct predictions, but also that they should have explanations on how customers will churn or not.

2.9 Research Gap Identification

Even though there has been remarkable success in the research on customer churn prediction, there are numerous gaps in research evident in the current literature. Firstly, the majority of the existing literature have conducted more research on the improvement of prediction accuracy

rather than on the understanding of the business and applicability to practice (Liu, Shengdong, et al., 2022). Although high prediction accuracy has been a priority, desirable insights are required by organizations in terms of factors that influence customer churn behavior.

Second, the research on the machine learning algorithms has been conducted on each case individually but comparative studies of various predictive algorithms in relation to the digital business environment with a subscription business model have been noticeably minimal. Hence, the field of different machine learning algorithms is not yet clear enough regarding the possible performance of each of them concerning the necessity to solve the problem of customer retention.

Third, bulk of the literature on customer churn research is on the conventional markets, such as telecommunication, banking and insurance. The subscription-based digital business model has received relative less attention despite the lack of mention of its increasing role in the world economy (Moch Yusuf Asyhari et al., 2025) . Subscriptions in digital businesses have distinctive features connected with customer behavior, consumption of services and customers that should be explored more. The other large gap in research has been on the analogy of feature importance and business interpretation.

Moreover, single evaluation measures, including accuracy, are often utilized in many studies, without any more comprehensive assessments of predictive performance. Various evaluation criteria such as precision, recall, F1-score and ROC-AUC should be employed to still achieve overall estimates of predicting ability (Moch Yusuf Asyhari et al., 2025).Finally, the contribution of machine learning has presented an opportunity of testing the new predictive algorithm options such as XGBoost and ensemble learning models on the churn environments of customers. Further studies are thus needed to assess the efficacy of these strategies as the study also looks at the implications of predictive analytics in real practice in customer retention management.

This paper will seal these gap areas in the research since it will offer a comparative analysis of different machine learning algorithm and will include the predictive performance analysis, feature exploration and business interpretation on the subscription based online business.

2.10 Conceptual Framework

The theoretical framework that will be developed on the basis of this research shows the relationship between the characteristic of customers, machine learning, predictive modeling,

and the customer churn outcomes. The framework is that customer demographic profile, customer behavioral variable, service usage variable and customer financial information all have an impact on customer retention behavior.

Some of the independent variables that will be considered in this study are the demographic variables such as gender, age based issues and family status, and customer dependency. List of varying behavior points is subscription time, usage patterns and customer contact indicators (Naga et al., 2025). Some of the service related variables are internet services, technical support, online security among other service subscriptions. These financial variables include charges per month, sum of money spent by the customers, and modes of payment.The variables undergo various steps of analytics, including data preprocess, feature engineering, exploratory data analysis, feature selection and machine learning models development. Four models of machine learning will be monitored and tested in the following steps: a Logistic Regression, a Decision Tree, a Random Forest, and an Extreme Gradient Boosting (XGBoost).

The attributes generated by these models are the predictive ones, therefore, customer churn detection and analyzing the feature significance (Naga et al., 2025). Finally, the conceptual framework underlines the creation of customer retention tactics, business decision-making, and predictive analytics tools in the digital business landscape with subscriptions.

CHAPTER 3: METHODOLOGY

3.1 Introduction

In this chapter, the approach to methodology that is going to be used to examine the concept of the customer churn prediction in the context of subscription-based online customers is introduced. The chapter describes the philosophy of the research project, the research approach, the research design, the characteristics of the dataset, feature selection, data preprocessing, data exploratory data analysis, machine learning algorithms, and the evaluation of model used in this research project. The research methodology was constructed in such a way to guarantee the reliability of research findings, their validity and repeatability and offer a full analysis of machine learning performance in customer churn prediction (Balla et al., 2025). The methodological strategy chosen provides an opportunity to conduct a predictive analysis as well as practical corporate inter pretation of the nature of customer retention behavior.

3.2 Research Philosophy and Approach

3.2.1 Research Philosophy

The current research assumes a positivist approach to research. Positivism presupposes that one can acquire knowledge by use of objective observation, empirical inquiry, statistical analysis, and scientific measurement. The positivist paradigm is a suitable philosophical basis of the study because customer churn prediction is a research with quantifiable customer factors, observable behavioral trends, and quantifiable business results.

Positivist method focuses on objectivity, reliability and evidence-based analysis, which makes it especially applicable in machine learning and predictive analytics research. The study employs the use of numerical customer data, statistical data analysis techniques in exploring the association between customer variables, and customer churn behavior. In addition, the research aims at establishing predictive models which can be objectively assessed and tested empirically with the help of developed performance metrics (Bhatnagar, 2025). Thus, positivist philosophy adoption would be certain to give the research results a systematic, measurable, and evidence-based business decision-making possibilities.

3.2.2 Research Approach

In this research, a quantitative research approach and a deductive research strategy is considered. Quantitative research is suitable, as the research aims at analyzing numerical client data, the prognostic accuracy, and the assessment of machine learning models deploying statistical methods. The quantitative method enables unbiased analysis and the formulation of predictive models that are credible.

Deductive research method was chosen, as the study is also based upon theoretic as well as empirical studies connected with the customer retention and churn behavior and predictive analytics topics. A literature review was used to first investigate existing knowledge on previous research. The machine learning then was created and put to tests with actual customer data on various real-life scenarios in order to test the relationships discovered in previous research (Bhatnagar and Srivastava, 2025). This method will allow testing the theoretical assumptions and adding more empirical information to the topic of predicting customer churn in subscription-based online companies.

3.3 Research Design

This paper adopts a quantitative experimental methodology of research that is grounded on supervised machine learning. The experimental research design is especially appropriate to conduct predictive analytics research since it allows developing systematic models, conducting controlled experiments, analyzing comparatively, and evaluating the performance objectively.

The study plan is composed of a number of steps. The initial phase consisted of customer churn data acquisition and exploration to learn how it could be configured, which variables it included, and what was the nature of its data. The second step was concerned with preprocessing data tasks, such as the treatment of missing values, feature transformation, encoding of category variables, as well as feature scaling (Bogaert & Delaere, 2023). These data cleaning steps made sure the data were competent to machine learning application.

Third phase entailed carrying out the exploratory data analysis to determine customer behavioral patterns, explore the relationship between variables, and comprehend customer retention nature. Visualization tools and statistical data were employed to analyze the customer demographics, subscription pattern, service utilization, and financial behavior.

The fourth step included the creation and training of four supervised machine learning models, which included among others, Logistic Regression, Decision Tree, Random Forest and Extreme Gradient Boosting (XGBoost). These algorithms are chosen according to their level of popularity, prediction ability, interpretability, and aptness to the customer churn prediction issue (Bogaert and Delaere, 2023).

The fifth phase involved the assessment and comparison of the model performance based on various classification metrics, such as accurate, precision, recall, F1-score, and ROC-AUC. Lastly, the analysis of the feature importance was done so that the customer characteristics would be identified that would have the largest contribution to the customer churn prediction.

The design of the experiments that will be used in this study makes it reproducible, transparent, objective, and applicable in practice and allows a thorough study of the performance of machine learning in the context of subscription-based digital business.

3.4 Dataset Description

3.4.1 Dataset Source

The dataset used in this research is the IBM Telco Customer Churn dataset that was retrieved on the Kaggle public data repository. This dataset has become well-established in academic publications and in data industry applications in the field of customer churn prediction and predictive analytics. The dataset has been chosen based on its relevance, accessibility, and the availability of all the customer details and also because it is suitable in the supervised machine learning analysis. Moreover, the data set will have customer demographic, behavioural, service-related and financial variables, which will be very relevant to the research agenda. The dataset can also be useful to binary classification problems due to the availability of customer churn labels.

3.4.2 Dataset Characteristics

IBM Telco Customer Churn dataset is a dataset which contains information of 7043 customers and has 21 variables. The data is comprised of demographic details, subscription details, utilization patterns, monetary variables and customer churn. Churn is the target variable, which is used to indicate whether the customers were under subscriptions or terminated.

The information provided can be both numeric and categorical. Examples of numerical variables consist of subscription tenure, monthly charges, total charges and customer demographic indicators. Some of the categorical variables are the types of contracts, modes of payment, internet services, technical support services, online security services and customer billing preferences.

Preliminary analysis of exploratory analysis showed that there were concealed missing values in TotalCharges variable, as well as class imbalance in the target variable. Therefore, preprocessing methods were used to enhance the quality of data and guarantee intelligent machine learning application (Boozary et al., 2025).

3.4.3 Dataset Variables

The dataset variables were grouped in four general categories namely: the demographic variables, the subscription variables, the service related variables and the financial variables. Demographical factors consisted of gender, being a senior citizen, partner and dependent status. Parameters that were associated with subscriptions were type of contract, tenure, mode of payment and billing.

The variables related to services were phone service, internet service, online security, online backup, device protection, technical support, streaming television and streaming movie subscriptions (Chajia and Nfaoui, 2024). The variables that were financial were the monthly charges and the amount of money that the customers spend.

The study dependent variable was the churn among customers, and it was a binary classification outcome where the customer opened or continued subscription (Chakrabarty and Sinha, 2025). The predictor variables have been chosen according to the results of past research, the relevance of the business, the result of the exploratory analysis, and other considerations of the machine learning features.

3.5 Variables and Feature Selection

The process of feature selection can be viewed as a crucial element of developing a machine learning model since the quality and relevance of input variables are crucial predictors of its predictive performance. The decisions concerning the feature selection in this study consisted of a combination of theoretical content, business relevancy, the results of exploratory analysis, and utilising machine learning methods of importance of a feature (Chang et al., 2024).

The dependent variable was customer churn in this study, which was a binary outcome of classification based on whether the customers would be retained or churned. Demographic factors, subscription details, customer service utilization rates, and financial features were selected as independent variables (Charizanis et al., 2025). Dependent variable chosen was senior citizen, which was a predictor variable, as well as partnership status, dependents, tenure, phone service, multiple lines, internet service type, online security, online backup, device protection, technical support, streaming television, streaming movies, contract type, paperless billing preference, payment methods, monthly charges, and total charges.

The exploratory data analysis also aided feature selection by establishing a good correlation between customer churn and different variables which included contract type, subscription tenure, monthly charges, internet service, and customer support services (DENISSE, 2025). Also, analysis of machine learning feature importance revealed that one of the strongest customer churn predictors was contract type, tenure, monthly charges, total charges, payment methods, online security, and technical support.

The predictive variables included in the search process also made sure that the predictive models used variables that bore high weight in customer churn prediction and at the same time

reduces model complexity and enhances predictive performance. The features chosen presented an all-encompassing depiction of customer behavior and subscription attributes in any digital business that is based on subscription.

3.6 Data Preprocessing

3.6.1 Missing Value Treatment

Preprocessing of the data began by detecting and filling gaps in the data. Though the dataset initially seemed complete, additional examination showed that there were eleven values that were hidden in the total charges variable that were missing. These metrics with unobservable data were transformed into numerical missing data with the relevant data transformations (DENISSE, 2025). Thereafter, median imputation methods were used to fill the missing, yet maintaining the statistics of the data. The use of median imputation was chosen due to the fact that it reduces the impact of the outliers, and it preserves the distributional characteristics of numerical variables. Such a preprocessing operation helped enhance quality data and make it compatible with machine learning algorithms.

3.6.2 Data Encoding

Categorical customer features were converted to numerical forms using the label encoding methods because the machine learning algorithms needed inputs as numerical values. Label encoding a numeric value was placed on categories of information that were categorical like contract type, method of payment, internet service, and customer support services. This metamorphosis allowed machine learning algorithms to handle customer information efficiently and keep categorical relations intact (Dr. ARIF ARFAT, 2023). The reason behind the choice of label encoding was that it was computationally simple and related to the machine learning algorithms applied in this study. After the processes of encoding, all variables were coded and transformed into numerical form and were predictable.

3.6.3 Feature Scaling

The feature scaling was allowed in order to make sure that disparities in measurement scales did not affect the performance of the machine learning models disproportionately. The transformations of numerical variables into the standardized distributions with a zero mean and unit variance, the StandardScaler technique, were applied to use the standardization procedures (Hasan, Zubair, et al., 2024) 1. The importance of feature scaling was especially noteworthy to algorithms like Logistic Regression, since variables whose numeric ranges are large can

dominatingly impact the model learning procedures. Standardization has made convergence of models, faster calculations and helped in creating more dependable predictive performance among various machine learning algorithms.

3.6.4 SMOTE Implementation

Customer churn dataset was highly skewed with about 73.5 percent of its customers in the non-churn grouping and 26.5 percent in the churn. In a bid to correct this imbalance, Training has been done on the Synthetic Minority Oversampling Technique (SMOTE) on the training set. SMOTE creates artificial observations to minority class data by interpolating between observed data. SMOTE usage enhanced the class representation and lowered model biasness towards the majority class predictions. This strategy increased predictive power, both in terms of customer at risk identification and overall predictive accuracy, of machine learning algorithms.

3.7 Exploratory Data Analysis Methods

The Exploratory Data Analysis (EDA) was used to comprehend customer behavioral trends, relationship among variables, data distributions, and in facilitating process features selections (Han, 2024). EDA is an important step in predictive analytics since it helps the researchers to create predictions about customer behavior and nature before they make the models.

In the process of the exploratory analysis, several visualization and statistical methods were used. The customer churn distributions and demographics were analyzed using frequency distribution and count plots. The numerical variables, such as the subscription tenure and the monthly fees, between the churn and non-churn groups of customers, were compared using boxplots (Gegeleso & Ebiesuwa, 2025). Bar charts were created to examine the associations between the churn of customers and the following categories of categorical variables namely-contract type, payment methods, and internet services. Also, correlation analysis was conducted through correlation matrices and heatmaps to analyze the relationship between numerical and encoded variables.

Essential patterns of customer behavior were revealed in the exploratory analysis. Month-to-month contracts showed significantly more customers had a higher churn rate, and those with an extended subscription tenure showed a more significant retention behavior. Equally, an increase in monthly fees and poor customer support services was related to a higher profile of customer churn.

3.8 Machine Learning Algorithms

3.8.1 Logistic Regression

Logistic Regression has been chosen as a control type of classification model because it is less complex and easy to interpret, as well as because it has been used in numerous studies related to customer churn prediction. The algorithm uses the relationships between predictor variables and binary customer outcomes to estimate the probability of customer churn (Imani, Joudaki, et al., 2025). The Logistic Regression offers interpretable coefficients and can be used to predict well where customer relationships are relatively linear in nature.

3.8.2 Decision Tree

The use of Decision Tree classification was due to its understandability and the fact that it produces hierarchical structures of decisions. Decision Trees categorize the observations based on the recursive partitioning of datasets based on decision rules as formed by the predictors. The algorithm allows visualizing the customer decision pathways and allows business interpreting customer churn behavior.

3.8.3 Random Forest

Random Forest has been chosen because of the high predictive validity, as well as its ability to mitigate overfitting by learning as a set of trees. The algorithm takes small random subsets of observations to build multiple decision trees and integrates predictions by voting (Imani, Joudaki, et al., 2025). Random Forest also has feature importance values, and it is thus especially suited to customer churn analysis.

3.8.4 XGBoost

Extreme Gradient Boosting (XGBoost) was used due to its high predictive accuracy, computing efficiency, and strength. XGBoost is based on the principles of gradient boosting, enhanced optimization algorithms and regularization processes. The algorithm has illustrated superior performance in a variety of customer churn prediction research and structured business courses.

3.9 Model Evaluation Techniques

Machine learning model evaluation is one of the most crucial phases of the predictive analytics research since the efficiency of the predictive models also has a direct impact on business decision-making and customer retention models. As customer churn prediction may be stated as a binary classification task, a set of evaluation measures were utilized to make sure that all model performance dimensions were assessed properly (Manzoor et al., 2024). The misleading

outcomes of using one test in the evaluation could be common especially with the analysis of disproportionate customer sets where most of the observations are on the retained customers and not on the churned customers.

The first measure of evaluation used in this analysis was accuracy the ratio of correctly classified observations to the number of observations. Precision gives a rough measure of predictive accuracy and the average classification ability. Nevertheless, due to the frequent imbalance in classes of customer churn datasets, accuracy alone might not give a full picture of model performance. This resulted in the inclusion of more performance measures as part of the performance plan.

To determine the accuracy of churn predictions, the accuracy of the predictions was gauged based on the proportion of customers who were predicted to churn, but did not cancel their subscription. Accuracy is especially essential in the context of businesses as large false-positive outcomes can lead to a wasteful treatment costs spent on customer retention, and improper resource distribution inside the organization (Maqbool & Qureshi, 2026). Models with high level of precision have more confidence on the use of targeting strategies on the customers.

In recall, also known as sensitivity, was used to determine the capacity of machine learning models to find real churn customers. Recall is one of the most paramount performance metrics in customer churn prediction since the inability to determine those customers who are going to exit a company can translate into substantial financial losses and loss of customer lifetime value. Good recall performance thus contribute to its active customer retention management and business sustainability (Imani et al., 2025).

F1-score had been used as a balanced measure of the model performance that consisted of combining precision and recall to form one performance measure. As customer retention decisions need to be made not only with correct prediction, but also with correct identification of customers, F1-score provides the more comprehensive analysis of predictive performance.

The Receiver Operating Characteristic Area Under the Curve (ROC-ASC) was also used to measure the discriminatory power of both models of machine learning. ROC-AUC assesses how predictive models do differentiate between churn and non-churn customers at different classification thresholds. The higher the ROC-AUCs, the better the classification and predictive ability.

Confusion matrices were created of each machine learning algorithm to help analyze and understand its performance further. The confusion matrix analysis allowed us to determine the true positives, true negatives, false positives and factual negatives thus facilitating the comparative analysis and business understanding (Nur, 2025). In addition, ROC curves were plotted to compare the performance of the machine learning models graphically and determine the best algorithm to use in predicting the customers to be churned.

3.10 Ethical Considerations

One of the most important aspects of research involving academic studies and predictive analytics is the role of ethical consideration since the obligation to have responsible data management, transparency, and protect privacy are inherent elements of research. The present research was written based on the accepted academic and ethical principles to guarantee the integrity, reliability and responsible use of machine learning methods in predicting customer churn.

The dataset used in this work was acquired in the IBM Telco Customer Churn dataset which can be accessed freely in the Kaggle repository to be used in the process of education, research and analysis (Saha et al., 2023). Completely anonymized customer data are all that are included in the dataset and there is no personally identifiable information of any kind of a customer in the dataset (names, addresses, telephone numbers, financial account details and other sensitive personal identifiers). As a result, the usage of this set of data did not generate any real threats concerning, personal privacy or confidentiality.

Principles of privacy of data were followed closely throughout the process of research. The research was not accompanied by the utilization of human data, interviews, questionnaires and experimentation. Thus, no formal ethical approval was needed with human subjects. However, any analysis performed was done in a responsible manner and in purely academic use without any intent to misuse the data.

The other crucial issue of ethics will regard responsible usage of machine learning algorithms and predictive analytics methods. Churn prediction of customers can affect business decision-making, treatment approach towards customers and resource distribution within an organization (Sana et al., 2024). Researchers and practitioners should, therefore, make sure that predictive models are created in a transparent manner, interpreted correctly and implemented in a fair manner. This work has thus highlighted significance of model interpretability using feature importance analysis and comparative performance analysis.

In predictive analytics research, algorithmic bias and fairness are other concerns. The machine learning models can unwillingly produce biased forecasts, when the training information has some hidden prejudice or disproportionate representation. Several metrics of evaluation were used as well in order to provide balanced model performance evaluation.

Moreover, the principles of academic integrity were followed to the letter. All the literature sources, theoretical frameworks and research findings were acknowledged and referenced accordingly (Torquato et al., 2026). All the machine learning models, analysis, interpretations, and conclusions in this research study were created with the use of educational and research purposes only. Thus, the study adds to responsible development of predictive analytics, customer relationship management and machine learning applications to the context of subscription-based digital businesses.

CHAPTER 4: RESULTS

4.1 Introduction

This chapter contains the findings achieved due to the utilization of the machine learning algorithms of customer churn prediction in digital business endeavors based on the subscription form of customer management. Analysis was conducted on the IBM Telco Customer Churn dataset which had 7,043 records of customers and different demographic, behavioural, service based and financial variables. Exploratory data analysis, research of the data and analysis of the customer behaviour are discussed in the chapter and then proceeded to creating a predictive model and testing it. This chapter will provide an opportunity to apply trends in customer retention, discuss the variables associated with customer churn and provide certain empirical evidence regarding customer behavior in the context of subscriptions to the digital space.

4.2 Dataset Overview

4.2.1 Dataset Structure

The IBM Telco Customer Churn dataset used to conduct the current research refers to the data on the customers of a telecommunication company in the form of subscription. The sample of data consists of 7,043 records with 21 variables (customer demographic characteristics, service-subscriptions, financial data and customer churn features).

The data have categorical and numerical variables. Categorical features are unique to customers due to the prevalence of categorical variables and financial data as well as period of

subscription are numeric and represented, respectively (Kaveh Faraji Googerdchi et al., 2024). The target variable is churn, a binary classification variable (outcome) that indicates whether the services were retained or not to the customers.

Table 4.1: Dataset Characteristics

Dataset Attribute	Value
Total Records	7,043
Total Variables	21
Predictor Variables	20
Target Variable	Churn
Numerical Variables	4
Categorical Variables	17
Missing Values	11
Classification Type	Binary

In the original dataset, there was missing data that was not observed in the TotalCharges variable, and had to be preprocessed before the model could be developed. Moreover, the data set possessed such characteristics as imbalanced classes and balancing methods needed to be implemented in the machine learning training process.

The range of various variables included in the dataset provides an adequate foundation to explore the trends of customer behavior and employ predictive analytics techniques to business operations based on a subscription system conduct.

```
df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')

print(df.head())

print("\nShape:")
print(df.shape)

print("\nData Types:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
```

Figure 4.1: Python Code for Dataset Import and Initial Exploration

```
Shape:
(7043, 21)

Data Types:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
None
```

Figure 4.2: Dataset Overview Output

```
Missing Values:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

Figure 4.3: Missing Values Analysis Output

4.2.2 Customer Churn Distribution

The customer churn distribution analysis indicated that it is very imbalanced with classes in the dataset. There were 5,174 customers (in total 7,043) who continued to use the service and 1869 customers who forgot to use it at all.

Table 4.2: Customer Churn Distribution

Customer Status	Frequency	Percentage
No Churn	5,174	73.46%
Churn	1,869	26.54%
Total	7,043	100%

These findings indicate that approximately three-quarters of the customers still remained subscribers and more customers also terminated their subscriptions. It is a real world distribution model which correctly represents what is happening in the real subscription business where churn usually represents a minority group of customers.

The unequal proportion of churn and non-churn customers has some key implications to the evolution of machine learning models. Biased trained classification algorithms are usually biased in classifying the customers, who are under threat of churning, and therefore they are less effective in identifying a customer that can be churned (Khan et al., 2025). In this way, the mechanisms of balancing such as Smaller Minority Oversampling Technique (SMOTE) were used when developing the models to improve the forecasting accuracy.

The Customer churn distribution also goes to show the usefulness of predictive customer analytics to empower, as even a relatively small amount of customers most likely to cancel their subscription can result in big business value.

4.3 Exploratory Data Analysis

4.3.1 Customer Churn Distribution Analysis

The introductory phase of exploratory data analysis would have been to analyse the customer churn results distribution and find the overall customer retention trends. Customer churn visualization revealed that there seems to be an unbalanced ratio between customer retention and churned customers with a majority of the customers retaining their subscriptions (Khan et al., 2025). It was discovered that 73.46% of the customers remained in the organization or only 26.54% of the customers had canceled their subscriptions. Although it would seem that the rates of customer retention could be considered really high, the list of a churned customer

remains quite impressive and is enough to trigger severe financial effects in case of the operations with a subscription business model.

This evidence can be used to show that customer turnover is a business issue, which is serious enough to be subject to proactive measures of business management. Moreover, the imbalance that is present in the dataset also points to the need to use various evaluation metrics as opposed to assessing the predictive model solely using classification accuracy (Khattak et al., 2023) . When it comes to business, even a minor enhancement of the retention of customers can generate big changes in profitability since the subscription revenue will be recurrent. Hence, the correct recognition of customers with high churn rates is an essential business goal.

The distribution analysis also proved that the data will have enough observations needed to apply supervised machine learning algorithms and to conduct valid predictive analytics investigations.

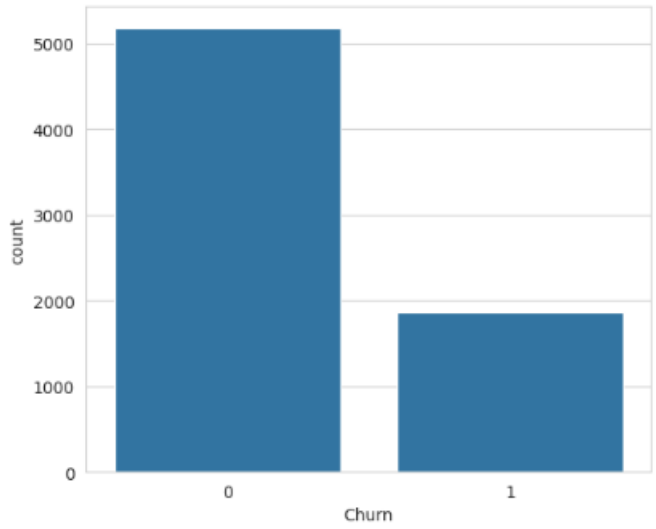


Figure 4.4: Customer Churn Distribution

4.3.2 Monthly Charges and Customer Churn Analysis

Monthly subscriptions were observed to find out their relationship with behavior of customers in churning. The boxplot analysis showed that there were huge differences in payments that were issued to the churned and retained customers in terms of the pattern of monthly payments. The cancellation rates were much higher among subscribers who had canceled their subscriptions as compared to those who remained with the subscriptions (Kosyk Daryna, 2024). The average monthly bill of the churned customers was approximately USD 80 and the average monthly bill of customers retained was approximately USD 64.

As the analysis shows, more delicate customers are ready to pay more with subscription fees: they are more sensitive to the service and price policies, other competitive options. Raising the subscription price may also end up heightening customer expectation regarding the performance of the service and customer care delivery quality (Lebelo et al., 2025). Besides, the greater the propensity of monthly payments observed among the churned customers means the higher the customer spending behavior variation among high-risk customer segment. This limit implies that the financial variables play an important role in customer retention.

At the managerial level, it is recommendable that organizations give a consideration to pricing strategy, service value proposition and customer satisfaction rates among the high paying customers to minimize the loss of customers and high retention in the long run.

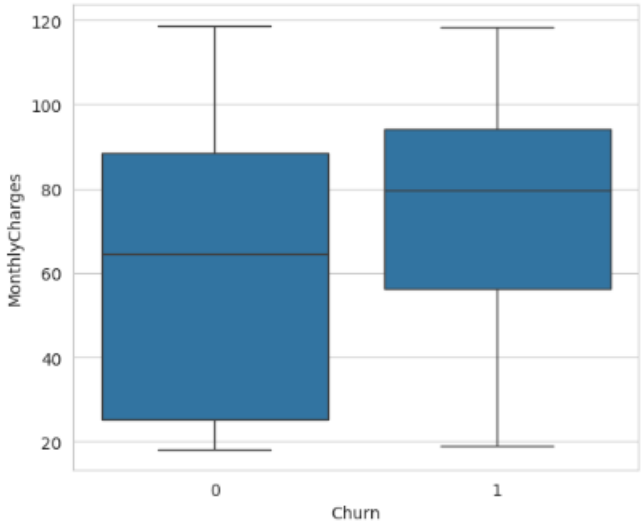


Figure 4.5: Monthly Charges versus Customer Churn Boxplot

4.3.3 Subscription Tenure and Customer Churn Analysis

One of the strongest relationships that were realized in the course of the exploratory analysis was through the subscription tenure analysis. Customers who continued to subscribe had significantly higher subscription rates as opposed to those who dropped their services (Li & Yan, 2025). The boxplot analysis revealed that the median subscription tenure of the churned customers was usually below 15 months and the retained customers had subscription tenure that were much higher with the majority being above 35 months. The findings indicate that customer attrition phenomenon is prevalent at the initial stages of customer relationship.

The immediacy of customer boarding, customer satisfaction with the service at the outset and customer engagement events form a long-term indicative of the customer retention process,

based on such observations. Customers able to pass initial stages of subscription appear to be much less willing to abandon their services (Liu, Ali, et al., 2022). The tenure of subscription is thus a very important predictor of customer loyalty and customer lifetime value. The findings can be used to suggest that during creation of customer relationship, businesses should prioritize on customer engagement schemes during initial periods of time in order to avoid chances of losing customers.

Close correlation, in turn, obtained between customer retention and the termination of subscription lends credence to the findings of previous researches and confirms the practicality of the customer lifecycle management in the environment of a business setup based on subscription.

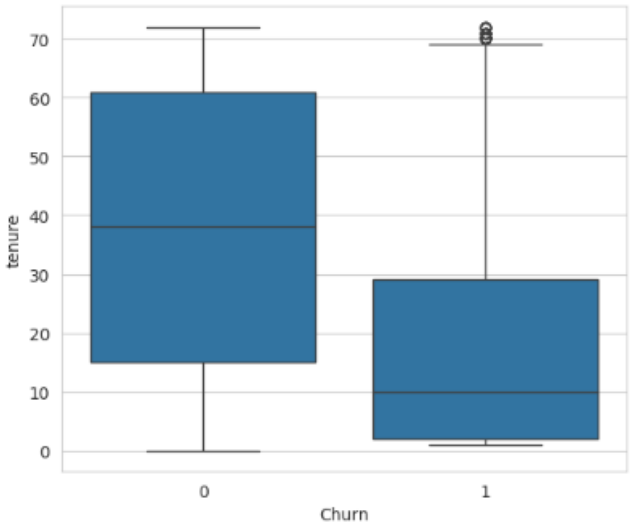


Figure 4.6: Subscription Tenure versus Customer Churn Boxplot

4.3.4 Contract Type and Customer Churn Analysis

In the exploratory analysis, the use of contract type emerged as one of the most effective predictors of the churn behavior of customers. The countplot analysis revealed that the patterns of customer retention with regard to the different contracts arrangements differed significantly, customers who subscribed to a contract of one month to month had extremely high churnage compared to those who had contracts of one year and two years (Liu et al., 2024). Among the churned customers, the majority of the customers were month to month clubbed but the long term contracted customers were much higherly retentive.

The results indicate that a contractual commitment is a key role in customer retention management. Customers who have a longer term contractual relationship appear to exhibit

reduced switching behaviour due to the financial obligation, switching costs and connection between the business and the customer (Maan & Maan, 2023). Further, the subscription deals in the long term enable an organization to enjoy better opportunities of improving customer engagement and upgrading service experiences. Negotiating customers, on the other hand, with the flexible contractual terms are freer to seek alternative sources of competition.

These results suggest that customer retention techniques to promote long term contractual agreements via loyalty rewards, price cuts, and additional service provisions should be pursued by organizations.

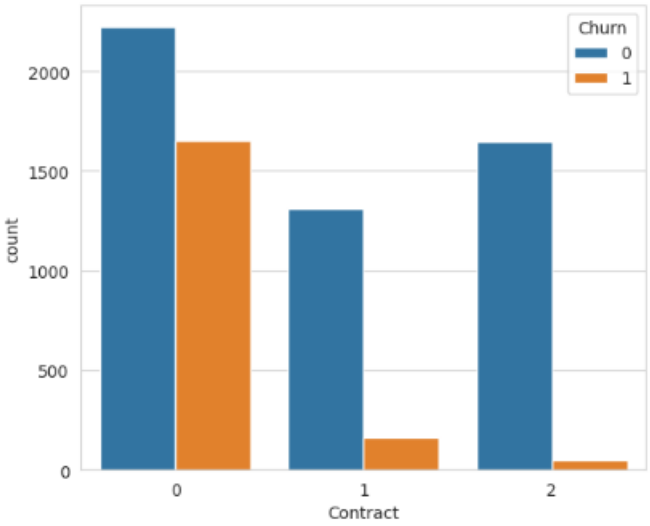


Figure 4.7: Contract Type versus Customer Churn Countplot

4.3.5 Internet Service and Customer Churn Analysis

A study of internet service type revealed that customers in different service groups had a very different behavior with respect to churning of their services. Churn rates were observed to be significantly higher in customers using fiber optic internet services in comparison to customers using DSL internet services or customers not subscribing to internet services (Manzoor et al., 2024). The results indicate that the customers who buy premium services might have higher service expectations and be more sensitive to the service quality, pricing policies, and competitive services. There also exists greater competition in the market and variety of services that fiber optic customers can have.

The customers who hadn’t canceled their account had quite low rates of customer churn, thus the customer churn-rates became synonymous with having relatively stable and contented

customers. The same applied to customers that were not subscribing to the internet services where the churn probability is relatively low.

The analysis of the correlation between the type of internet service and customer services retention shows the importance of quality management of the services offered and alignment of customer experience. Firms in the high end segments have to concentrate on customer satisfaction programs and customer support intervention programs in order that the firm is able to minimize the losses of customers (Maqbool and Qureshi, 2026). The analysis also indicates that service related variables are significant predictors of customer churn would be of great importance and has to be given a high regard when coming up with the predictive model.

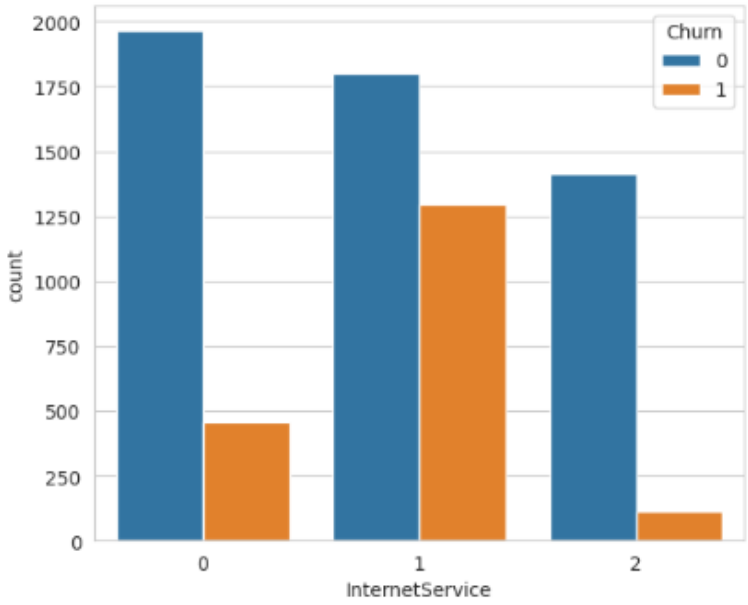


Figure 4.8: Internet Service versus Customer Churn Countplot

4.3.6 Correlation Analysis

Correlation was used to examine the relationship between the numerical and encoded variables in the customer churn data set. To determine any significant relationships between customer characteristics and customer churn behavior, a heatmap of correlation was created. The tenure of subscription as seen in the analysis had one of the strongest negative relationships with customer churn indicating that the chances of customers canceling their subscriptions are negatively associated with the tenure of the services (Matuszelański & Kopczewska, 2022). Going on the same tempo, the type of contract was significantly negatively correlated with the churn behavior of the customers.

Monthly charges were positively linked to customer churn meaning that those whose charges are higher in comparison to their subscriptions are likely to cancel. The overall customer spending was positively correlated with the time of tenure to a subscription since the amount of money spent on customers was cumulative with time (Muhammed Kotan et al., 2025). The correlation analysis also formed moderate relationships form the different service related variables and this implies the occurrence of interactions among customer service preferences and customer retention behavior. However, it did not show any indications of strong multicollinearity amid the predictor variables.

These results conclude that the churn behavior of customers is a result of a complicated interaction between demographic, financial, service, and behavioral variables and, therefore, it is beneficial to use advanced machine learning algorithms that are able to model nonlinear associations.

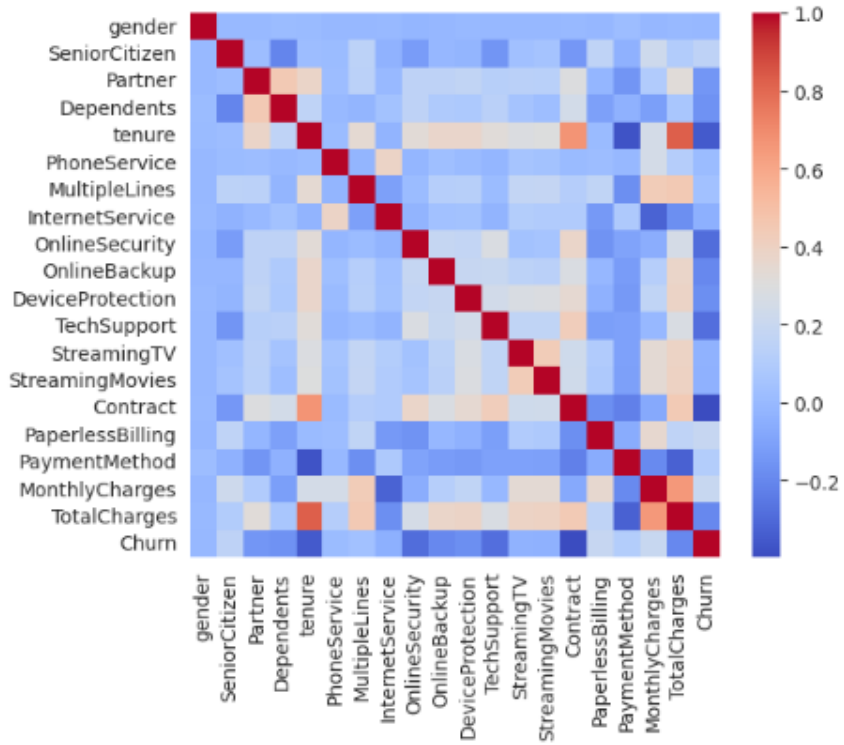


Figure 4.9: Correlation Heatmap

4.4 Data Preprocessing Results

4.4.1 Data Cleaning and Transformation

Several preprocessing procedures were performed before running machine learning algorithms in an attempt to improve the quality of data and their appropriateness in predictive

model/objectives. The initial step was to examine the dataset structure whereby gaps in data were identified and parameters transformed into the correct format of analysis.

The customer Identification variable (CustomerID) was eliminated as it was a unique variable and not predictive to customer churn classification. Subsequently, TotalCharges variable was converted to a numerical format following the string form then transformed it back to a numerical form following appropriate data conversion procedures (Murat Basal et al., 2025). In the process of transformation, eleven missing values which were not visible were determined. In order to limit the loss of information to the highest point, the missing numbers were filled with median imputation, and with the purpose of maintaining the statistical distribution of the variable.

Categorical data was coded with methods of label encoding after cleaning data. This operation was possible to transform variables such as type of contract, mode of payment; internet services and nature of customer care to numerical numbers, which can be interpreted by machines. After completing encoding steps, the StandardScaler algorithm was exploited to normalize feature values so that to normalize numerical variables and optimize the performance of the model (Natarajan, 2023). The data were then divided into training and test parts 80:20 respectively. The sample used in training was 5,634 customers and in testing, the sample was 1,409 customers. Such preprocessing operations made the dataset appropriate in terms of selecting supervised machine learning algorithms and the comparative performance assessment.

Table 4.3: Data Preprocessing Summary

Activity	**Result**
CustomerID Removal	Completed
TotalCharges Conversion	Completed
Missing Values Treated	11
Label Encoding	Applied
Feature Scaling	Applied
Train-Test Split	80:20
Training Samples	5,634
Testing Samples	1,409

```
X = df.drop('Churn', axis=1)
y = df['Churn']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# scaling
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# handle imbalance
smote = SMOTE(random_state=42)

X_train, y_train = smote.fit_resample(
    X_train,
    y_train
)

print(X_train.shape)
print(y_train.shape)
```

Figure 4.10: Python Code for Data Cleaning and Feature Engineering

4.4.2 Class Balancing Using SMOTE

The exploratory analysis indicated that the customer churn dataset had a huge imbalance in terms of classes. Specifically, it had only 26.54% observations in churn (and 73.46% in non-churn). Such imbalance can frequently be linked to predictive bias wherein machine learning models, which forecast better those observations of those in the majority group compared to the minority group customers (Nur, 2025).

The SMOTE (Synthetic Minority Oversampling Technique) on the training set was used to address this problem. SMOTE can be used to create artificial observations within the sample of the minority classes to generate some perceptive neighbors that enhance class balance, reducing this bias in the predictive.

The training set of 4139 non-churn and 1495 churn customers was used to implement the SMOTE. Oversampling caused an equal distribution of the customers in the resulting training data to reach 8,278 and the customers in each category (n=4,139).

Table 4.4: Class Distribution Before and After SMOTE

Class	Before SMOTE	After SMOTE

No Churn	4,139	4,139
Churn	1,495	4,139
Total	5,634	8,278

The use of SMOTE greatly enhanced the capability of machine learning algorithms in the identification of risky customers to churn. This preprocessing step played a significant role in better recall results and predictive power of all the models used.

```
(8278, 19)
(8278,)
```

Figure 4.11: SMOTE Implementation Output

4.5 Logistic Regression Analysis

4.5.1 Model Development

Logistic Regression was chosen as the baseline model of classification because it is simple, interpretable and popular in the research studies on customer churn. The model was trained on the balanced training dataset created based on implementing SMOTE. The binary target variable was the customer churn status, standardized customer features were used as predictor variables (Omari et al., 2025).

The Logistic Regression algorithm estimates the churn behavior of each of the customers by use of the logistic functions that estimate the relationship between the customer characteristics and the customers responding to churning. The model is built using default hyperparameter and its performance is evaluated to build the predictive ability using numerous performance measures.

The biggest advantage of Logistic Regression is its interpretability that enables organizations to comprehend the impacts of various characteristics of customers on their churn chances (Ong et al., 2024). The model thus becomes an important point of reference when they compare more sophisticated methods of machine learning.

4.5.2 Model Performance Analysis

The Logistic Regression model showed high predictive performance, and high results on various evaluation measures.

Table 4.5: Logistic Regression Performance Results

Metric	Value
Accuracy	74.24%

Precision	50.95%
Recall	78.88%
F1-Score	61.91%
ROC-AUC	83.90%

The model had a general classification of 74.24%. What is more important, this model showed a remarkably high level of recall, being 78.88%, which means it is highly effective in removing the number of customers who terminated their subscriptions.

The results of the confusion matrix analysis showed that the model was very accurate in predicting 751 retained customers and 295 churn customers. However, 284 retained customers were erroneously discovered as churn customers whereas 79 churn customers were not seen.

Table 4.6: Logistic Regression Confusion Matrix

Actual/Predicted	No Churn	Churn
No Churn	751	284
Churn	79	295

The ROC-AUC 83.90% is the maximum level value of the model, hence, depicting the best performance due to its maximum level of discrimination. This observation implies that the Logistic Regression is a good way of differentiating between churn and non-churn customers (Ong et al., 2024).

In business terms, the high levels of recall of Logistic Regression make it of high value in customer retention management since it is usually more significant to find customers with high churn rates as opposed to lowering false positive based on the classification.

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Logistic Regression	0.742370	0.509499	0.788770	0.619098	0.838916

Figure 4.13: Logistic Regression Performance Output

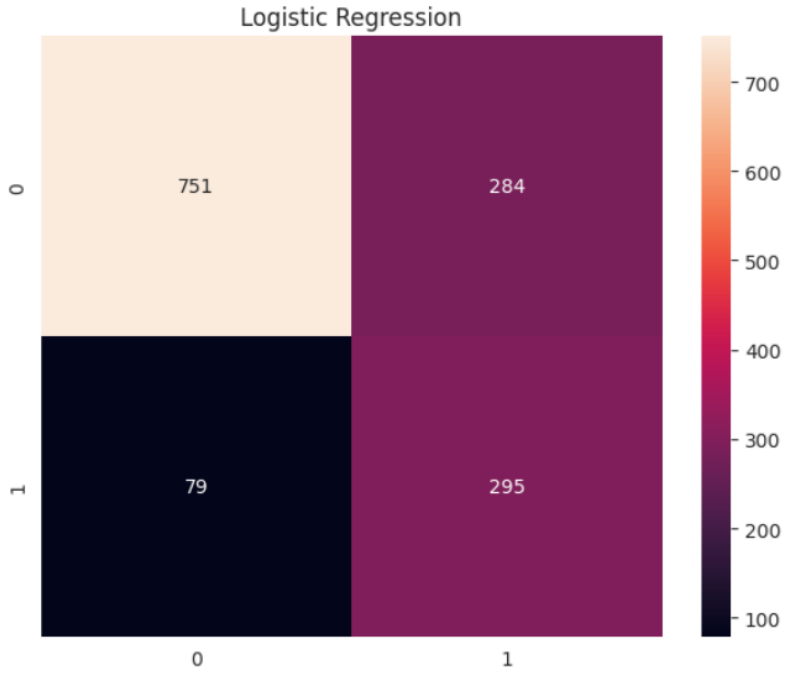


Figure 4.14: Logistic Regression Confusion Matrix

4.6 Decision Tree Analysis

4.6.1 Model Development

The Decision Tree classification implementation was undertaken to explore predictive potential of tree-based classification framework in predicting customer churn. The algorithm recursively divides the observed customers into hierarchical customer classification trees based on decision rules estimated (depending on predictor variables). Decision Trees bear significant benefits in terms of interpretation since they offer explicit decision-making frameworks that could assist organizations in knowing customer classifications paths (Özalpay & Taşkın, 2024). More so, Decision Trees do not need complicated assumptions about the data distributions to formulate an effective process of both numerical and categorical customer variables.

Decision Tree model was trained on the balanced training dataset and tested on various performance measures to determine how well the model predicts and its application in business.

4.6.2 Model Performance Analysis

The weakest predictive performance of the machine learning algorithms assessed was the Decision Tree model.

Table 4.7: Decision Tree Performance Results

Metric	Value

Accuracy	71.89%
Precision	47.49%
Recall	55.61%
F1-Score	51.23%
ROC-AUC	66.71%

The model had an accuracy of 71.89% that was the lowest compared to other tested models. The recall (at 55.61%) is also fairly good in discriminating against potentially churning customers.

The confusion matrix analysis revealed that the model was able to identify 805 retained customers and 208 churn customers correctly. The model, nevertheless, failed to find 166 customers who eventually became churners.

Table 4.8: Decision Tree Confusion Matrix

Actual/Predicted	**No Churn**	**Churn**
No Churn	805	230
Churn	166	208

This ROC-AUC (66.71%) value additionally indicates that there is quite low discriminatory power in respect to the other models.

The poorer performance of the Decision Tree model could have been explained by the tendency to overfitting as well as the lack of the capacity to depict unusual customer behavior patterns (Peng et al., 2023). But, the use of Decision Trees is not entirely in vain as this is a simple type that can be interpreted and realized in business.

```
1        Decision Tree  0.718950   0.474886  0.556150  0.512315  0.666749
```

Figure 4.17: Decision Tree Performance Output



Figure 4.18: Decision Tree Confusion Matrix

4.7 Random Forest Analysis

4.7.1 Model Development

Random Forest classification has been used as a method of learning in an ensemble to overcome the weaknesses of single decision trees. The algorithm builds a series of decision trees based on random sets of observations and predictor variables and then integrates the predictions they make via majority voting methods (Roberts et al., 2022).

Random Forest was chosen because it has a good predictive power, robustness and has the capability of modelling complex nonlinear relationships. In addition, the algorithm also generates the scores of feature importance that can be understood by the business and locate customer behavior. The balanced training dataset was used to train the model, and the model was tested in terms of various classification metrics to identify if it was suitable in predicting customer churn.

4.7.2 Model Performance Analysis

Random Forest consistently presented high predictive ability and had balanced results in classification.

Table 4.9: Random Forest Performance Results

Metric	Value

Accuracy	76.93%
Precision	56.22%
Recall	58.02%
F1-Score	57.11%
ROC-AUC	82.22%

The model achieved accuracy score of 76.93% which was incredibly better than the other two models namely, the Logistic Regression and Decision Tree. Balanced precision and recall effect was also exhibited by the model.

As demonstrated by the confusion analysis method, the Random Forest was able to make predictions with a total of 867 and 217 retained customers and churn customers, respectively.

Table 4.10: Random Forest Confusion Matrix

Actual/Predicted	No Churn	Churn
No Churn	867	168
Churn	157	217

The ROC-AUC of 82.22% represents a high level of discriminatory and shows that the model can be used effectively to draw the line between churn and non-churn customers.

The high work of Random Forest is explained by an ensemble organization that decreases the variation and night to a minimum and enhances generalization power (Saha et al., 2023). Also, the Random Forest represents well the multifaceted interactions among customers.

These results suggest that ensemble learning methods offer a major benefit in terms of customer churn prediction and customer retention analytics.

```
2        Random Forest  0.768630   0.562176  0.580214  0.571053  0.821835
```

Figure 4.21: Random Forest Performance Output

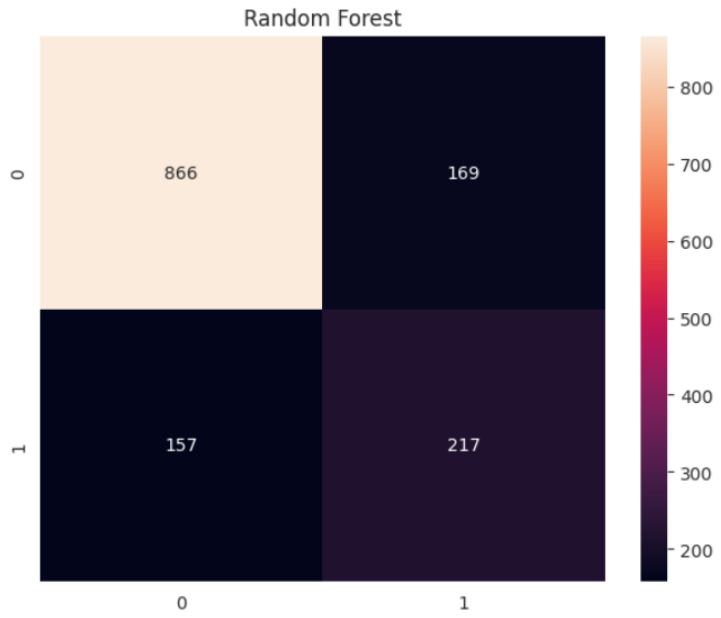


Figure 4.22: Random Forest Confusion Matrix

4.8 XGBoost Analysis

4.8.1 Model Development

Extreme Gradient Boosting (XGBoost) was committed because of its impressive predictability, performance efficiency and ability to acquire intricacy of customer behavioral patterns. XGBoost is based on the principles of gradient boosting and supplemented with highly developed optimization and regularization algorithms (Sana et al., 2022).

The model used the balanced training dataset to train the model and to optimize the model to gain more predictive accuracy and reduce overfitting (Sana et al., 2024). XGBoost was selected since it has been demonstrated to be successful in predictive analytics study and customer churn prediction solutions.

4.8.2 Model Performance Analysis

Among all evaluated algorithms, XGBoost demonstrated the strongest overall predictive performance.

Table 4.11: XGBoost Performance Results

Metric	Value
Accuracy	77.50%
Precision	57.25%
Recall	60.16%
F1-Score	58.67%

ROC-AUC	81.44%

The accuracy of 77.50% recorded in the model was the best with respect to classification ability.

The message using the confusion matrix analysis indicated that the model was also correct in detecting 867 and 225 retained customers and churn customers respectively.

Table 4.12: XGBoost Confusion Matrix

Actual/Predicted	No Churn	Churn
No Churn	867	168
Churn	149	225

The high predictive power of the model was once again supported by the value of the ROC-AUC 81.44%.

This may be attributed to the superiority of XGBoost that comes with the capacity to capture nonlinear relationships, enhanced prediction errors courtesy of an iterative procedure and model decreasing variance (Sanches et al., 2025). XGBoost has the best predictive, recall and classification reliability balance in business.

```
3              XGBoost  0.775018   0.572519  0.601604  0.586701  0.814246
```

Figure 4.25: XGBoost Performance Output

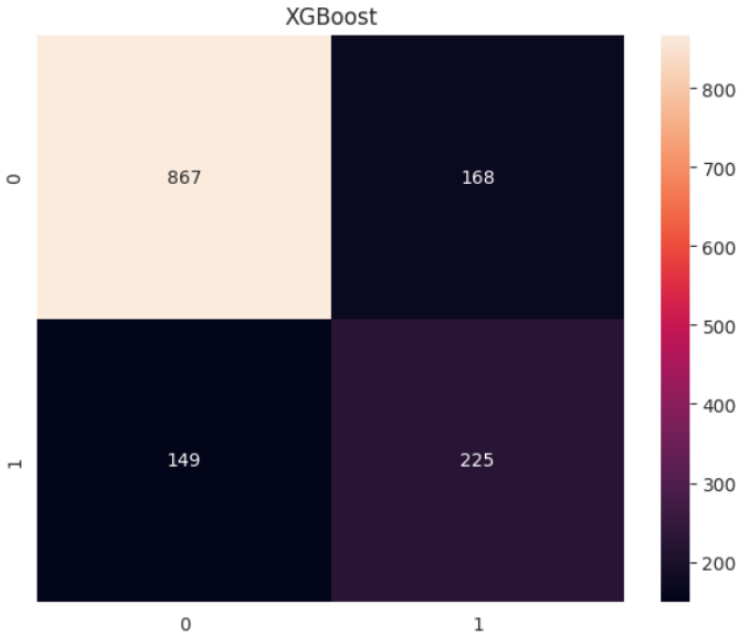


Figure 4.26: XGBoost Confusion Matrix

4.9 Model Performance Comparison

4.9.1 Comparative Performance Analysis

Table 4.13: Comparative Performance Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	74.24%	50.95%	78.88%	61.91%	83.90%
Decision Tree	71.89%	47.49%	55.61%	51.23%	66.71%
Random Forest	76.93%	56.22%	58.02%	57.11%	82.22%
XGBoost	77.50%	57.25%	60.16%	58.67%	81.44%

The results of comparison point out that XGBoost was the most accurate, whereas the most recall and ROC-AUC were achieved by Logistic Regression. It was determined that Decision Tree ranked the lowest based on the predictive performance of both random forest and XGBoost; both these have balanced prediction.

4.9.2 ROC Curve Comparison

The comparison of the ROC curves also proved the better performance of ensemble learning methods. The Logistic Regression provided the highest ROC-AUC of 83.90%, random Forest (82.22%) followed and XGBoost (81.44%) provided lower ROC-AUC. Decision Tree had the lowest discriminatory capability.

The results are that, even with more advanced machine learning algorithms, simpler statistical models have the potential to give very good customer identification results.

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Logistic Regression	0.742370	0.509499	0.788770	0.619098	0.838916
1	Decision Tree	0.718950	0.474886	0.556150	0.512315	0.666749
2	Random Forest	0.768630	0.562176	0.580214	0.571053	0.821835
3	XGBoost	0.775018	0.572519	0.601604	0.586701	0.814246

Figure 4.28: Model Performance Comparison Table Output

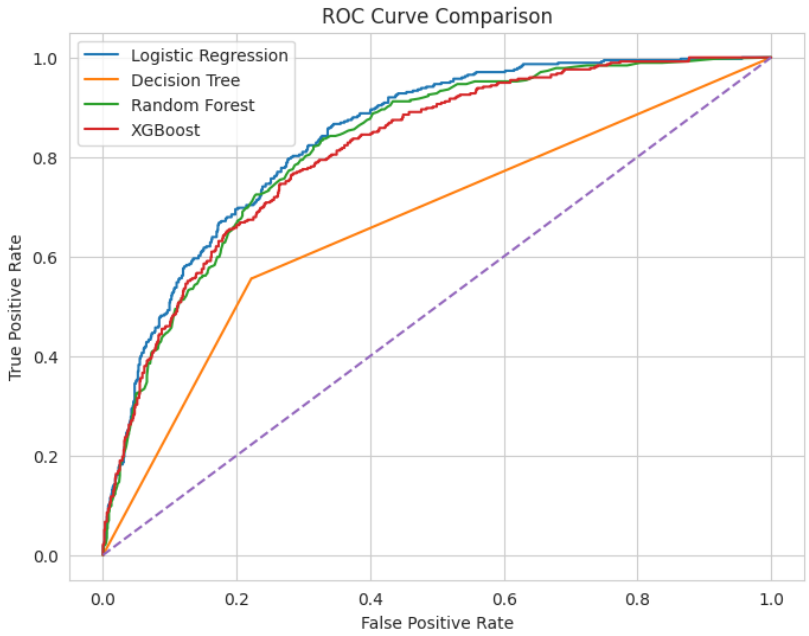


Figure 4.30: ROC Curve Comparison Graph

4.10 Feature Importance Analysis

4.10.1 Feature Ranking Analysis

Random forest algorithm was applied to extract the features that had maximum customer churn prediction and plotted the results using the feature importance tools.

Table 4.14: Top Feature Importance Rankings

Rank	Feature	Importance
1	Contract	0.146
2	Tenure	0.136
3	MonthlyCharges	0.133
4	TotalCharges	0.132
5	PaymentMethod	0.076
6	OnlineSecurity	0.064
7	TechSupport	0.050
8	InternetService	0.037
9	PaperlessBilling	0.035
10	OnlineBackup	0.028

The discussion showed that contract type was the best predictor of customer churn with subscription tenure and monthly charges being very close.

4.10.2 Business Interpretation of Feature Importance

The feature importance analysis has some valuable information to use in customer retention management. The pre-eminence of the contract type shows that the customers that have the

short-term contracts are more likely to be cancelled. Similarly, tenure of subscription depicts the fact that customer loyal customers increase over time.

The total expenditure and monthly charges show that pricing processes have a significant part in the behaviour of customer retention. Besides, the payment options relevance, online security service, and support services indicate that customer satisfaction and service quality will also play a vital role in customer retention as well (Sina Mirabdolbaghi and Amiri, 2022).

The summary of implications these findings are presented as follows: companies ought to target the customer engagement strategy over the long-term, more advanced customer services, enhanced pricing mechanisms, and active customer retention schemes of those customers groups who can be named as being at risk.

```
rf = results[
    'Random Forest'
]['model']

importance = pd.DataFrame({

    'Feature':X.columns,

    'Importance':
        rf.feature_importances_

})

importance = importance.sort_values(
    by='Importance',
    ascending=False
)

plt.figure(figsize=(8,8))

sns.barplot(

    x='Importance',

    y='Feature',

    data=importance.head(10)
)

plt.title(
    'Top 10 Important Features'
)

plt.show()

print(
    importance.head(10)
)
```

Figure 4.31: Feature Importance Python Code

	Feature	Importance
14	Contract	0.146594
4	tenure	0.136079
17	MonthlyCharges	0.133613
18	TotalCharges	0.131727
16	PaymentMethod	0.075572
8	OnlineSecurity	0.064281
11	TechSupport	0.050294
7	InternetService	0.036620
15	PaperlessBilling	0.034531
9	OnlineBackup	0.028112

Figure 4.32: Feature Importance Output

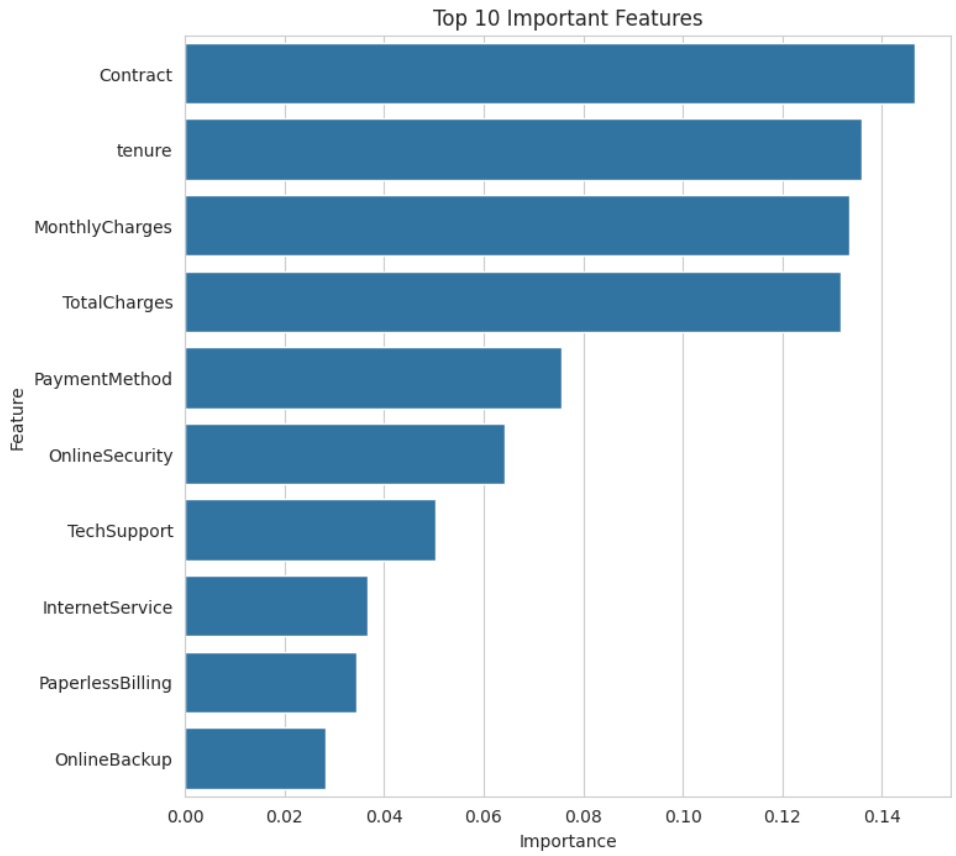


Figure 4.33: Top 10 Feature Importance Graph

CHAPTER 5: CONCLUSION

5.1 Introduction

In the chapter, the discussion and interpretation of the findings obtained along the research on machine learning methods in churn prediction of customers in the digital business of subscription will be given. The chapter evaluates identified relationships on the approaches of the exploratory analysis, the business relevance of the feature importance results is addressed

and the performance of machine learning algorithms compared. In addition, the chapter also compares and contrasts with the existing academic literature and underlines practical implications of predictive analytics to customer retention management. The results discussed in this chapter not only make a contribution to the theory but also practical observations of how machine learning techniques are implemented in customer relationship management and business decision-making.

5.2 Discussion of Findings

5.2.1 Customer Churn Behavior Analysis

The results of this research indicate that the interaction of demographic, behavioral, service-related, and financial factors affects customer churn behavior in subscription-based digital business in a rather complicated setting. The exploratory analysis revealed that there are several important characteristics of the customers that are important in winning the customers and, therefore, validating the multidimensionality of customer retention behavior.

Some of the major findings of this study include the association between the length of subscription and customer retention. The analysis indicated that a relatively low subscription history customers have a higher probability to churn as compared to long-term customers. The implication of this result is that customer loyalty is built up over an extended period and that initial customer experiences are important factors that can be relied on to forecast long-term retention rates (Theodoridis & Tsadiras, 2022). The trend towards longer subscription seems to develop relationships with service providers and allows the customers to be more committed to the service provider and reduce switching at the cost of customer commitment.

One of the main lessons is that there is a correlation between monthly subscription fee and churn of customers. There were higher chances of termination among the customers whose monthly payment was more. This observation is a manifestation that pricing can significantly influence the customer decision-making process and the customer with a higher financial commit or requirement might have a more substantial expectation of the quality of services offered and customer support (Torquato et al., 2026).

The analysis, further revealed that contract type is assumed to be one of the most promising predictors of customer retention behavior. The rates of customer churn on customers in the month to month contractual agreements were higher than on customers in the long term

contracts. The conclusion is that customer commitment mechanisms and contractual incentives are important in customer retention management.

Further, attributes that conveyed details about the services offered such as type of internet services, online security services and availability of technical support were found to be highly connected with customer retention outcomes. These results imply that customer experience and service quality is still a critical factor in customer loyalty in the subscription-based business setting.

Altogether, the customer behavior analysis proves that customer churn prediction needs an in-depth grasp of the personality of the customers, their behavior styles, and the business relationship.

5.2.2 Machine Learning Performance Analysis

Comparison of machine learning algorithms indicated that there are remarkable discrepancies between the forecasting performances of the frameworks granted. The results show that machine learning methods can help to predict the churn of the customers and would be effective to help manage the effort related to customer retention.

XGBoost model was also good at predicting since it scored the highest as a whole with the score of 77.50%. The success of XGBoost might be due to the fact that it tries to approximate nonlinear relationships, its prediction error is optimized iteratively, and its variability is also regularized (via regularization mechanisms) (Tuğçe Aydin Hataş et al., 2023). The presence of the balance performance observed in the case of accuracy, precision, recall and F1-score also testify that XGBoost is a good tradeoff between predictive and business applicability.

Random Forest was also great at predicting the outcome as the accuracy score was 76.93% and ROC-AUC was 82.22%. Random Forest has a robust ensemble structure which helped it to have a good generalization capability. Furthermore, valuable feature importance measures were also provided by the algorithm, which enhanced business knowledge.

The final accuracy of the Logistic Regression was lower than the overall accuracy of the ensemble learning procedures however, the highest recall (78.88%) and highest ROC-AUC figuration (83.90%) were obtained. These results suggest that Logistic Regression can be used effectively to identify customer at risk of churn as this is one of the most important needs in customer retention management (Tuğçe Aydın Hataş et al., 2023). This points to the potential important influence of easier predictive models on practical utility even with the rise of more

complex machine learning algorithms, as indicated by the high recall rates by Logistic Regression.

The lowest predictive performance was with Decision Tree classification, which had an accuracy score of 71.89% and a ROC-AUC of 66.71%. Such poor results of the Decision Tree model as compared to the rest can be explained by the overfitting effects, and its inabilities to cover the complex customer relationships.

On the whole, the results support the idea that ensemble learning methods give significant benefits to the customer churn prediction and prove the significance of the choice of predictive models depending on the specific business aims.

5.2.3 Feature Importance Interpretation

The feature importance analysis presented valuable information about the customer features that are most useful to predict customer churn. The findings indicated that the predictors of the retention behavior by the customer are ideal where the kind of contract, the term of subscription, monthly payment and the total payment, security services offered by the online shop and the availability of the technical support.

The position of the subscription-based business model mechanisms of customer commitment is accentuated by the fact that the type of contract is regarded the most important predictor. When the customer signed a long term contract, the likelihood of churn was very low as compared to when the customer signed a short term contract as a subscription. The implication of this observation is that firms should encourage customer loyalty programs and contractual benefits to enhance customer relationships on a long-term basis.

The second most important predictive was subscription tenure because it reflects the importance of customer relationship building and customer lifecycle management. The respective critical impact of monthly charges and consumer expenditure in general also provides confirmation on the applicability of pricing strategies and the perceived worth of the client in the customer retention process (Usman Gani Joy et al., 2024).

Additionally, the importance of technical support and online security services implies that the experience of the customers using the services and its quality influence customer satisfaction and customer loyalty significantly. Those companies, who want to improve their customer retention strategy, are advised to therefore infrastructural invest in the field of customer support

services, reliability of services offered to customers and customer experience management programs.

5.3 Comparison with Previous Studies

The findings obtained in this research have a great coincidence with the earlier literature research on study of customer churn prediction and use of machine-learning in customer relationship management. The results not only affirm many of these known relations in the previous studies on customer churn, but also present novel data as to the comparative effectiveness of machine-learning techniques.

The applicability of subscription tenure which was discovered in the current study fits quite well into the findings of other literatures on customer retentions. Numerous researches have shown over and over again that the more customers are already served the higher the chances of them turning into loyal and hence not dropping the service (Ramesh et al., 2024). This conclusion is also supported by the current research as it demonstrates that the subscription tenure is among the strongest predictors of customer retention in the digital business environment that revolves around subscription.

Similarly, the fact that the type of contract has been named the most interesting predictor of customer churn to deliver the results can be compared with those reported in the telecommunications, software services and customer churn types of business literature publications. Previous studies have repeatedly established that customers whose operations are believed to be subjected to the arrangement of long-term contracts exhibit less switching behavior and more customer loyalty in comparison with the customers, who subscribe to a flexible arrangement.

The findings on the monthly charges and customer expenditure are also similar to what is being researched on the importance of pricing policies and customer perceived value. Earlier studies have overwhelmingly shown that the financial variables have a substantial impact on customer retention behaviour, especially in well-competitive subscription markets.

The operational performance analysis, in the comparison of the machine learning algorithms also depicts very similar outcomes with the past researches. Previous literature has recorded that ensemble learning algorithms such as the Random Forest and gradient boosting algorithms have the advantage of being more effective compared to traditional approaches to statistics and single decision trees (Rasool et al., 2025)). Improvement in the results of XGBoost was

enormous in this study again explaining the growing literature upon the effectiveness of gradient boosting techniques in customers churn prediction.

However, an interesting finding that emerges out of this study is in terms of the performance of Logistic Regression. Even though more advanced machine learning processes are becoming prevalent, the greatest recall and ROC-AUC rates were achieved with the use of Logistic Regression as compared to the remaining models compared. This observation suggests that the classic predictive models can nevertheless be incredibly useful in customer retention aspects particularly when the needs are high such as identifying customers rather than overall classification performance.

The existing study is also relevant to the current existing body of work with respect to the reality of it combining the importance of features with a business context. Although much of the earlier studies have been concerned with the predictive accuracy, this study shows the practical significance of knowing the factors that influence the behaviour of customers to churn (Prakash, 2024). The results consequently add to the current body of knowledge regarding the adoption of machine learning techniques in the context of subscription-based online companies and explain the necessity to continue adopting predictive analytics in the customer relationship management processes.

5.4 Practical Implications

The observation of the present study can have a number of key practical implications to the subscription-based digital companies, aiming at enhancing the customer retention and decreasing the customer attrition rates, and increasing the overall effectiveness of the company in the long-term. The achievements of machine learning algorithms in churning the customer have proven that predictive analytics have the power to be a useful decision support tool in customer relationship management and strategic business planning.

One of the most practical implications is the knowledge of the capacity of organizations to recognize the high-risk customers before the loss of customers. The predictive models produced in this paper and particularly XGBoost, Random Forest and Logistic Regression, had a great capacity to predict customers who were likely to cancel their subscriptions (Prakash, 2024). This predictive nature allows the organizations to have proactive retention tactics as opposed to reactive customer recovery tactics.

The results also highlight the significance of customer segmentation and retention efforts that are individualized. Since the variables affecting the customer churn were type of contract, tenure of subscription, monthly fee and quality of service offered, an organization can develop custom interventions programs to a specific group of customers. Customers with personalized offers or month to month contracts and shorter histories of subscriptions should be offered discounted long subscriptions, e.g.

Moreover, the significance of technical assistance and online security services outlined in this study indicates that the quality of investments in customer service might work wonders concerning customer retention rates (Naga et al., 2025) . To improve customer loyalty business must put in consideration the larger role in improving the customer experience, response in customer service and management of customer experience.

The results also indicate predictive analytics potentially supports the strategic decision-making of pricing policies, service design, customer engagement programs, and resource distribution. Embedded machine learning systems on customer relationship management platforms would enable organizations to continuously monitor the customer behavioural trends, and make retention decisions based on the trends.

5.5 Limitations of the Study

Although this research provides the right information on how to predict customer churning through machine learning techniques, there are several limitations, which should be identified. Firstly, the researcher selected a single publicly available data that relied upon the IBM Telco Customer Churn repository (Naga et al., 2025). This is a famous data set that has been extensively used in the literature on the customer churn but this may not be a comprehensive data that is applicable in a digital business model which is founded on a subscription fee. Therefore, the applicability of the results can be questionable.

Second, the research limited itself to structured customer information, such as demographic variable, behavioral variables, service related variables, and financial variables. Other potentially significant information about customers that were not incorporated in the predictive models include customer sentiments, customer feedback, and customer interactions via social media (Moch Yusuf Asyhari et al., 2025).

Third, the study only examined 4 supervised machine learning algorithms such as Logistic Regression, the decision tree, the random forest and the XGBoost. Although they are amongst

the most common predictive algorithms, other machine learning systems like deep learning algorithms, and hybrid predictive systems were not studied.

Lastly, the research took a quasi-historic forecast technique using past descriptive information of customers instead of exploring up to date customer behavior prediction software. In this way, future research may expand the customer churn prediction areas by introducing new data, customer variables and advanced methods of analysis.

5.6 Reflection on the Research Process

This research experience was a learning experience, which provided me invaluable academic, technical, and analytical skills in machine learning, predictive analytics, and customer relationship management. The detailed course of the research resulted in a large amount of knowledge on how machine learning practices can be implemented into a concrete business environment.

The literature review stage provided a balanced opinion of the churn behavior of the customer, subscription business model, predictive analytics and machine learning methodology. The literature review helped realize the complexity of the customer retention management process and mention the risk of enhanced importance of data-grounded decision-making in contemporary companies.

Practical implementation phase was big in terms of constructing technical skills associated with the preprocessing of data, exploratory data analysis, feature engineering, train machine learning models and predictive performance metrics. Data cleaning, class imbalance, predictive algorithms, and machine learning interpretation were all helpful experience in applying data science in practice.

In addition, comparative analysis of the machine learning algorithms demonstrated that predictive performance is not only usually hinged about the selection of the algorithms but hinged as well about the quality of the data, feature engineering, preprocessing techniques and business insights. The importance of the necessity to combine both technical analytic dexterity along with the practicalizing company knowledge were also reaffirmed by the research.

5.7 Future Research Directions

Future research could build upon this study by exploring other applications of machine learning and artificial intelligence, such as deep learning, neural networks, reinforcement learning, and hybrid predictive models. Addiction of customer sentiment analysis, social media information,

customer feedback information and real time behavioural analytics can be added as well to boost predictive value among researcher. In addition, future research can focus on customer churn prediction in a variety of industries and subscription sites to enhance the extrapolability of study results (Moch Yusuf Asyhari et al., 2025). Explaining models of artificial intelligence and real-time implementation of a customer retention system also presents a substantial aspect of future research to customer analytics.

References

AbdelAziz, N. M., Bekheet, M., Salah, A., El-Saber, N., & AbdelMoneim, W. T. (2025). A Comprehensive Evaluation of Machine Learning and Deep Learning Models for Churn Prediction. *Information*, *16*(7), 537. https://doi.org/10.3390/info16070537

Adiputra, I. N. M., Wanchai, P., & Lin, P.-C. (2025). Optimized customer churn prediction using tabular generative adversarial network (GAN)-based hybrid sampling method and cost-sensitive learning. *PeerJ Computer Science*, *11*, e2949. https://doi.org/10.7717/peerj-cs.2949

Agrawal, S. (2025). Machine Learning Methods for Churn Prediction and Infrastructure Resilience. *Mit.edu*. https://hdl.handle.net/1721.1/162934

AL-Najjar, D., Al-Rousan, N., & AL-Najjar, H. (2022). Machine Learning to Develop Credit Card Customer Churn Prediction. *Journal of Theoretical and Applied Electronic Commerce Research*, *17*(4), 1529–1542. https://doi.org/10.3390/jtaer17040077

Alam, M. M., Prithul, S. S., & Hossain, Md. S. (2026). Customer Behavior, Adoption and Retention in Software Markets: An AI-Based Analysis for Strategic Growth of Software Business. *Lecture Notes in Networks and Systems*, 29–43. https://doi.org/10.1007/978-3-032-13003-7_4

Aljifri, A. (2024). *Predicting Customer Churn in a Subscription-Based E-Commerce Platform Using Machine Learning Techniques*. DIVA. https://www.diva-portal.org/smash/record.jsf?pid=diva2:1857189

Altairey, H. A., & Al-Alawi, A. I. (2024). Customer Churn Prediction in the Telecommunication Industry Using Multiple Machine Learning Algorithms. *2024 International Conference on Open Innovation and Digital Transformation (OIDT)*, 1–10. https://doi.org/10.1109/oidt59407.2024.11082687

Asif, D., Arif, M. S., & Mukheimer, A. (2025). A data-driven approach with explainable artificial intelligence for customer churn prediction in the telecommunications industry. *Results in Engineering*, *26*, 104629. https://doi.org/10.1016/j.rineng.2025.104629

Ayodeji O.J. Ibitoye, Oluwaseun Kolade, & Olufade F.W. Onifade. (2025). Customer retention model using machine learning for improved user-centric quality of experience through personalised quality of service. *Journal of Business Analytics*, 1–19. https://doi.org/10.1080/2573234x.2025.2551950

Babatunde, R., Olaniyi Abdulsalam, S., Abdulkabir Abdulsalam, O., & Olaolu Arowolo, M. (2023). Classification of customer churn prediction model for telecommunication industry using analysis of variance. *IAES International Journal of Artificial Intelligence (IJ-AI)*, *12*(3), 1323. https://doi.org/10.11591/ijai.v12.i3.pp1323-1329

Balla, R. T., Bhandari, R. S., & Kumar, V. (2025). Survival Analysis for Estimating Customer Churn in Subscription-Based Digital Platforms. *2025 International Conference on Computational Engineering, Sensing Technology and Management (ICCETM)*, 1–5. https://doi.org/10.1109/iccetm66557.2025.11557777

Bhatnagar, A. (2025). Customer Churn Prediction using Machine Learning Approach: A Comprehensive Study. *Journal of Information Systems Engineering & Management*, *10*(25s), 80–92. https://doi.org/10.52783/jisem.v10i25s.3944

Bhatnagar, A., & Srivastava, S. (2025). Customer Churn Prediction: A Machine Learning Approach with Data Balancing for Telecom Industry. *International Journal of Computing*, 9–18. https://doi.org/10.47839/ijc.24.1.3873

Bogaert, M., & Delaere, L. (2023). Ensemble Methods in Customer Churn Prediction: A Comparative Analysis of the State-of-the-Art. *Mathematics*, *11*(5), 1137. https://doi.org/10.3390/math11051137

Boozary, P., Sheykhan, S., GhorbanTanhaei, H., & Magazzino, C. (2025). Enhancing customer retention with machine learning: A comparative analysis of ensemble models for accurate churn prediction. *International Journal of Information Management Data Insights*, *5*(1), 100331. https://doi.org/10.1016/j.jjimei.2025.100331

Chajia, M., & Nfaoui, E. H. (2024). Customer Churn Prediction Approach Based on LLM Embeddings and Logistic Regression. *Future Internet*, *16*(12), 453. https://doi.org/10.3390/fi16120453

Chakrabarty, P., & Sinha, R. (2025). A Hybrid Deep Learning and XGBoost Framework for Predicting OTT Subscription Timing Based on User Behavior Analysis. *2025 IEEE 4th World Conference on Applied Intelligence and Computing (AIC)*, 25–30. https://doi.org/10.1109/aic66080.2025.11212176

Chang, V., Hall, K., Xu, Q. A., Amao, F. O., Ganatra, M. A., & Benson, V. (2024). Prediction of Customer Churn Behavior in the Telecommunication Industry Using Machine Learning Models. *Algorithms*, *17*(6), 231–231. https://doi.org/10.3390/a17060231

Charizanis, G., Mavridou, E., Vrochidou, E., Kalampokas, T., & Papakostas, G. A. (2025). Data-Driven Decision Support in SaaS Cloud-Based Service Models. *Applied Sciences*, *15*(12), 6508. https://doi.org/10.3390/app15126508

Dang Tran, H., Le, N., & Nguyen, V.-H. (2023). Customer Churn Prediction in the Banking Sector Using Machine Learning-Based Classification Models. *Interdisciplinary Journal of Information, Knowledge, and Management*, *18*, 087-105. https://doi.org/10.28945/5086

DENISSE, C. (2025). Predictive Modeling of Customer Churn in the Home-to-Home Program Industry: A Time-to-Churn Analysis. *Unive.it*. https://hdl.handle.net/20.500.14247/25035

Dr. ARIF ARFAT. (2023). HCL TECHNOLOGIES LIMITED: AN INVESTIGATION INTO THE FINANCIAL PERFORMANCE ANALYSIS OF THE COMPANY. *International Journal of Data Science and IoT Management System*, *2*(4), 32–36. https://doi.org/10.64751/

Fakhar Bilal, S., Ali Almazroi, A., Bashir, S., Hassan Khan, F., & Ali Almazroi, A. (2022). An ensemble based approach using a combination of clustering and classification algorithms to enhance customer churn prediction in telecom industry. *PeerJ Computer Science*, *8*, e854. https://doi.org/10.7717/peerj-cs.854

Gegeleso, B., & Ebiesuwa, O. (2025). Comparative Analysis of Random Forest and LSTM Models for Customer Churn Prediction Based on Customer Satisfaction and Retention. *Indonesian Journal of Data and Science*, *6*(2), 313–323. https://doi.org/10.56705/ijodas.v6i2.244

Han, S. (2024). Machine Learning Based Customer Churn Prediction in Banking Sector. *Highlights in Business Economics and Management*, *40*, 378–384. https://doi.org/10.54097/7z0s8b66

Hasan, B., Zubair, Shaikh, S. A., Khaliq, A., & Nadeem, G. (2024). Data-Driven Decision-Making: Accurate Customer Churn Prediction with Cat-Boost. *The Asian Bulletin of Big Data Management*, *4*(02), Science 4(2)–250. https://doi.org/10.62019/abbdm.v4i02.175

Hasan, M. S., Siam, M. A., Ahad, M. A., Hossain, M. N., Ridoy, M. H., Rabbi, MD. N. S., Hossain, A., & Jakir, T. (2024). Predictive Analytics for Customer Retention: Machine Learning Models to Analyze and Mitigate Churn in E-Commerce Platforms.

Journal of Business and Management Studies, *6*(4), 304–320. https://doi.org/10.32996/jbms.2024.6.4.22

Imani, M., Joudaki, M., Beikmohamadi, A., & Arabnia, H. R. (2025). *Customer Churn Prediction: A Review of Recent Advances, Trends, and Challenges in Conventional Machine Learning and Deep Learning*. https://doi.org/10.20944/preprints202503.1969.v1

Imani, M., Majid Joudaki, Beikmohammadi, A., & Hamid Reza Arabnia. (2025). Customer Churn Prediction: A Systematic Review of Recent Advances, Trends, and Challenges in Machine Learning and Deep Learning. *Machine Learning and Knowledge Extraction*, *7*(3), 105–105. https://doi.org/10.3390/make7030105

Kathi Chandra Mouli, Raghavendran, C. V., Bharadwaj, V. Y., Vybhavi, G. Y., C. Sravani, Khristina Maksudovna Vafaeva, Rajesh Deorari, & Hussein, L. (2024). An analysis on classification models for customer churn prediction. *Cogent Engineering*, *11*(1). https://doi.org/10.1080/23311916.2024.2378877

Kaveh Faraji Googerdchi, Asadi, S., & Seyed Mohammadbagher Jafari. (2024). Customer churn modeling in telecommunication using a novel multi-objective evolutionary clustering-based ensemble learning. *PLoS ONE*, *19*(6), e0303881–e0303881. https://doi.org/10.1371/journal.pone.0303881

Khan, M. A., Keshari Samal, S., Sharma, G., Singh, A., & Khan, A. (2025). Machine Learning for Predictive Customer Retention in Competitive Markets. *2025 International Conference on Computational Engineering, Sensing Technology and Management (ICCETM)*, 1–6. https://doi.org/10.1109/iccetm66557.2025.11557887

Khattak, A., Mehak, Z., Ahmad, H., Asghar, M. U., Asghar, M. Z., & Khan, A. (2023). Customer churn prediction using composite deep learning technique. *Scientific Reports*, *13*(1), 17294. https://doi.org/10.1038/s41598-023-44396-w

Kosyk Daryna. (2024). *Customer retention rate forecasting in subscription-based business models in the mobile app domain*. Ucu.edu.ua. https://er.ucu.edu.ua/items/f8560b6d-5909-4d2a-bf4c-4753938144fd

Lebelo, M., Moeti, M. N., & Langa, M. R. (2025). A Deep Learning-Based Online Subscription Customer Segmentation and Retention Prediction Model for E-Commerce Platforms. *2025 5th International Multidisciplinary Information Technology and Engineering Conference (IMITEC)*, 1–6. https://doi.org/10.1109/imitec67386.2025.11410463

Li, Y., & Yan, K. (2025). Prediction of bank credit customers churn based on machine learning and interpretability analysis. *Data Science in Finance and Economics*, *5*(1), 19–35. https://doi.org/10.3934/dsfe.2025002

Liu, R., Ali, S., Bilal, S. F., Sakhawat, Z., Imran, A., Almuhaimeed, A., Alzahrani, A., & Sun, G. (2022). An Intelligent Hybrid Scheme for Customer Churn Prediction Integrating Clustering and Classification Algorithms. *Applied Sciences*, *12*(18), 9355. https://doi.org/10.3390/app12189355

Liu, X., Xia, G., Zhang, X., Ma, W., & Yu, C. (2024). Customer churn prediction model based on hybrid neural networks. *Scientific Reports*, *14*(1). https://doi.org/10.1038/s41598-024-79603-9

Liu, Y., Shengdong, M., Jijian, G., & Nedjah, N. (2022). Intelligent Prediction of Customer Churn with a Fused Attentional Deep Learning Model. *Mathematics*, *10*(24), 4733. https://doi.org/10.3390/math10244733

Maan, J., & Maan, H. (2023). *Customer Churn Prediction Model using Explainable Machine Learning*. ArXiv.org. https://arxiv.org/abs/2303.00960

Manzoor, A., Qureshi, M. A., Kidney, E., & Longo, L. (2024). A review on machine learning methods for customer churn prediction and recommendations for business practitioners. *IEEE Access*, *12*, 1–1. https://doi.org/10.1109/access.2024.3402092

Maqbool, M., & Qureshi, I. H. (2026). A systematic literature review on explainable AI for churn prediction, customer segmentation and retention. *International Journal of Data Science and Analytics*, *22*(1). https://doi.org/10.1007/s41060-025-01018-0

Matuszelański, K., & Kopczewska, K. (2022). Customer Churn in Retail E-Commerce Business: Spatial and Machine Learning Approach. *Journal of Theoretical and Applied Electronic Commerce Research*, *17*(1), 165–198. https://doi.org/10.3390/jtaer17010009

Moch Yusuf Asyhari, Pratiwi Susanti, & Yessi Yunitasari. (2025). Implementation of Machine Learning Models for Predicting Internet Service Provider Customer Churn. *Journal of Computer Networks Architecture and High Performance Computing*, *7*(4). https://doi.org/10.47709/cnahpc.v7i4.7012

Muhammed Kotan, Ömer Faruk Seymen, Levent Çallı, Kasım, S., Yavuz, B. Ç., & Tijen Över Özçelik. (2025). A novel methodological approach to SaaS churn prediction using whale optimization algorithm. *PLoS ONE*, *20*(5), e0319998–e0319998. https://doi.org/10.1371/journal.pone.0319998

Murat Basal, Khouloud Moulai, & Anıl Cetin. (2025). Predictive Analytics for Customer Behavior Prediction in Artificial Intelligence. *Economics World*, *12*(2). https://doi.org/10.17265/2328-7144/2025.02.006

Naga, V., Suvaris, R. D., Divya Nimma, & Bala, B. K. (2025). Advanced AI-Driven Predictive Modeling for Enhancing Customer Retention in Subscription-Based Service Platforms. *Lecture Notes in Electrical Engineering*, 449–463. https://doi.org/10.1007/978-981-96-9203-3_37

Natarajan, L. (2023). Facilitating Customers towards Product Subscription in Fin-Tech Application through Behavioural Analysis - NORMA@NCI Library. *Ncirl.ie*. https://norma.ncirl.ie/7576/1/lakshmirajnatarajan.pdf

Nur, M. (2025). ADVANCEMENTS IN MACHINE LEARNING FOR CUSTOMER RETENTION: A SYSTEMATIC LITERATURE REVIEW OF PREDICTIVE MODELS AND CHURN ANALYSIS. *Journal of Sustainable Development and Policy*, *01*(01), 250–284. https://doi.org/10.63125/9b316w70

Omari, A., Al-Omari, O., Al-Omari, T., & Fati, S. M. (2025). A predictive analytics approach to improve telecom's customer retention. *Frontiers in Artificial Intelligence*, *8*. https://doi.org/10.3389/frai.2025.1600357

Ong, J.-X., Tong, G.-K., Khor, K.-C., & Haw, S.-C. (2024). Enhancing Customer Churn Prediction With Resampling: A Comparative Study. *TEM Journal*, 1927–1936. https://doi.org/10.18421/tem133-20

Özalpay, G., & Taşkın, N. (2024). Effective Churn Prediction in the Online Payment Sector: A Survival Analysis. *CONF-IRM 2024 Proceedings*. https://aisel.aisnet.org/confirm2024/13/

Peng, K., Peng, Y., & Li, W. (2023). Research on customer churn prediction and model interpretability analysis. *PLOS ONE*, *18*(12), e0289724–e0289724. https://doi.org/10.1371/journal.pone.0289724

Prakash, R. (2024). Customer Churn Prediction in Retail Banking using Predictive Analytics. *Ncirl.ie*. https://norma.ncirl.ie/8777/1/rahulprakash.pdf

Ramesh, P., R. Nithyanandhan, M.K.Mohamed Faizal, V. Nivakumar, M. Nalini, & Siva Subramanian R. (2024). *Machine Learning Strategies for Customer Churn Prediction in Competitive Enterprises*. 653–660. https://doi.org/10.1109/icscna63714.2024.10864143

Rasool, H. A., Aljawaheri, K. K., & Karram, A. A. M. (2025). Explainable Customer Churn Prediction in Telecom Using Ensemble Learning and SHAP Analysis. *Mathematical Modelling of Engineering Problems*, *12*(12), 4373–4386. https://doi.org/10.18280/mmep.121226

Roberts, M., Ignacio, D. J., Ihshaish, H., & Zhu, Y. (2022). *Estimating defection in subscription-type markets: empirical analysis from the scholarly publishing industry*. ArXiv.org. https://arxiv.org/abs/2211.09970

Saha, L., Tripathy, H. K., Gaber, T., El-Gohary, H., & El-kenawy, E.-S. M. (2023). Deep Churn Prediction Method for Telecommunication Industry. *Sustainability*, *15*(5), 4543. https://doi.org/10.3390/su15054543

Sana, J. K., Abedin, M. Z., Rahman, M. S., & Rahman, M. S. (2022). A novel customer churn prediction model for the telecommunication industry using data transformation methods and feature selection. *PLOS ONE*, *17*(12), e0278095. https://doi.org/10.1371/journal.pone.0278095

Sana, J. K., Sohel, R. M., & Saifur, R. M. (2024). *Privacy-Preserving Customer Churn Prediction Model in the Context of Telecommunication Industry*. ArXiv.org. https://arxiv.org/abs/2411.01447

Sanches, H. E., Ayslan Trevizan Possebom, & Ruiz, B. (2025). Churn prediction for SaaS company with machine learning. *Innovation & Management Review*. https://doi.org/10.1108/inmr-06-2023-0101

Shaikhsurab, M. A., & Magadum, P. (2024). *Enhancing Customer Churn Prediction in Telecommunications: An Adaptive Ensemble Learning Approach*. ArXiv.org. https://arxiv.org/abs/2408.16284

Siddiqui, N., Md Asraful Haque, Khan, S., Adil, M., & Shoaib, H. (2024). Different ML-based strategies for customer churn prediction in banking sector. *Journal of Data Information and Management*. https://doi.org/10.1007/s42488-024-00126-z

Sina Mirabdolbaghi, S. M., & Amiri, B. (2022). Model Optimization Analysis of Customer Churn Prediction Using Machine Learning Algorithms with Focus on Feature Reductions. *Discrete Dynamics in Nature and Society*, *2022*, 1–20. https://doi.org/10.1155/2022/5134356

Suh, Y. (2023). Machine learning based customer churn prediction in home appliance rental business. *Journal of Big Data*, *10*(1). https://doi.org/10.1186/s40537-023-00721-8

Theodoridis, G., & Tsadiras, A. (2022). Applying machine learning techniques to predict and explain subscriber churn of an online drug information platform. *Neural Computing and Applications*. https://doi.org/10.1007/s00521-022-07603-9

Torquato, E. L., de Melo, G. A., Peixoto, M. G. M., Nogueira, T. H., de Sousa Pereira, M., Serrano, A. L. M., & Fogaça, R. R. (2026). Analysis of recurring revenue models in subscription-based software companies. *Future Business Journal*, *12*(1). https://doi.org/10.1186/s43093-026-00809-6

Tuğçe Aydın Hataş, Emir Obalı, Ayşenur Yıldız, Sibel Kırmızıgül Çalışkan, Veysel Karani Yılmaz, Kara, E., Yasemin Kurtçu Meşe, Ebru Öz, & Tuna Çakar. (2023). *Analyzing Customer Churn: A Comparative Study of Machine Learning Models on Pay-TV Subscribers in Turkey*. *3*, 1–6. https://doi.org/10.1109/iisec59749.2023.10390998

Usman Gani Joy, Kazi Ekramul Hoque, Mohammed Nazim Uddin, Chowdhury, L., & Park, S.-B. (2024). A Big Data-Driven Hybrid Model for Enhancing Streaming Service Customer Retention through Churn Prediction Integrated with Explainable AI. *IEEE Access*, *12*, 69130–69150. https://doi.org/10.1109/access.2024.3401247

Xiahou, X., & Harada, Y. (2022). B2C E-Commerce Customer Churn Prediction Based on K-Means and SVM. *Journal of Theoretical and Applied Electronic Commerce Research*, *17*(2), 458–475. https://doi.org/10.3390/jtaer17020024

Zhang, T., Moro, S., & Ramos, R. F. (2022). A Data-Driven Approach to Improve Customer Churn Prediction Based on Telecom Customer Segmentation. *Future Internet*, *14*(3), 94. https://doi.org/10.3390/fi14030094

Zhou, Y., Chen, W., Sun, X., & Yang, D. (2023). Early warning of telecom enterprise customer churn based on ensemble learning. *PloS One*, *18*(10), e0292466–e0292466. https://doi.org/10.1371/journal.pone.0292466

Appendix

Colab link:

https://colab.research.google.com/drive/1ZBDXR8wOR3reTGsiZkQfqZ2Gq4UB2Lk-?usp=sharing

Kaggle link: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

```
rf = results[
    'Random Forest'
]['model']

importance = pd.DataFrame({

    'Feature':X.columns,

    'Importance':
        rf.feature_importances_

})

importance = importance.sort_values(
    by='Importance',
    ascending=False
)

plt.figure(figsize=(8,8))

sns.barplot(

    x='Importance',

    y='Feature',

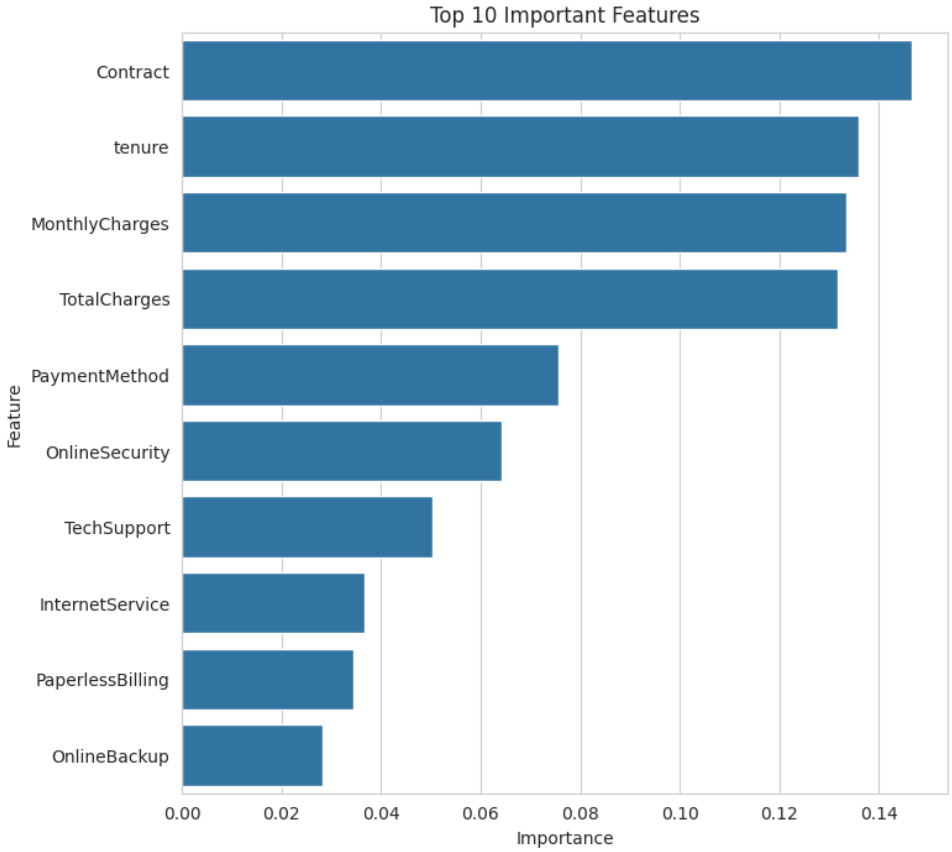
    data=importance.head(10)
)

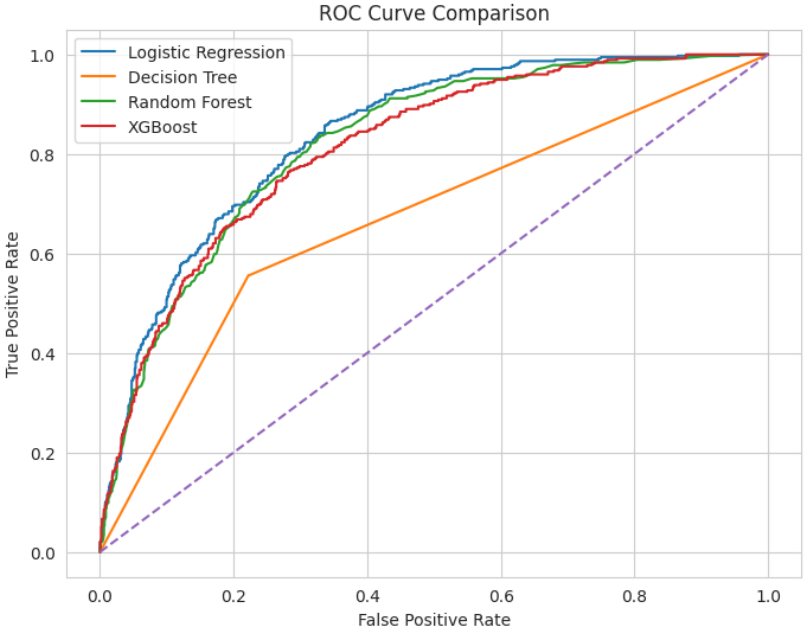
plt.title(
    'Top 10 Important Features'
)

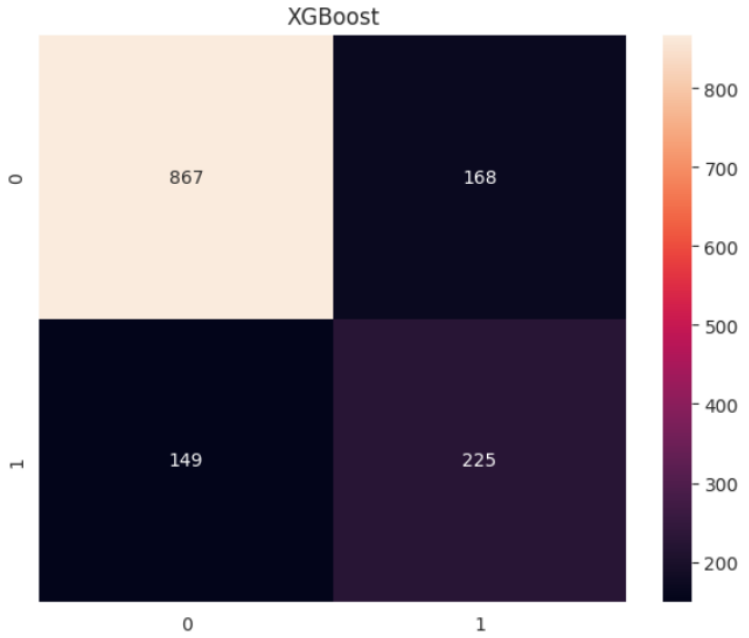
plt.show()

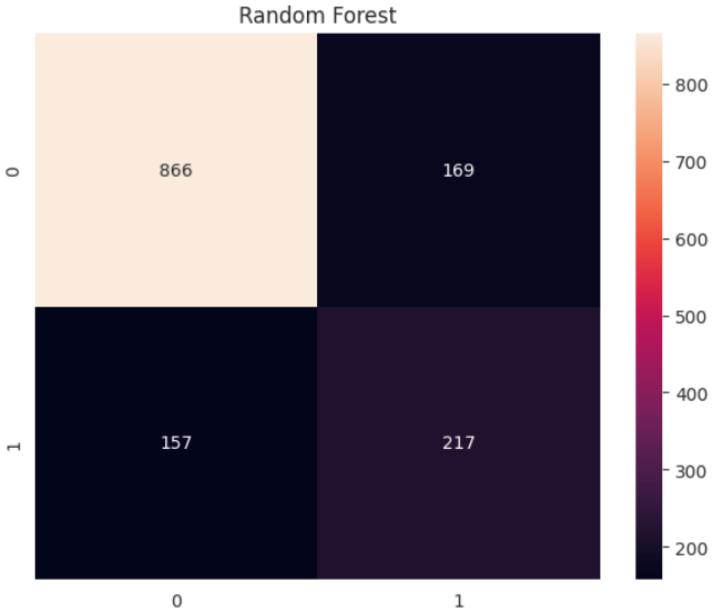
print(
    importance.head(10)
)
```

	Feature	Importance
14	Contract	0.146594
4	tenure	0.136079
17	MonthlyCharges	0.133613
18	TotalCharges	0.131727
16	PaymentMethod	0.075572
8	OnlineSecurity	0.064281
11	TechSupport	0.050294
7	InternetService	0.036620
15	PaperlessBilling	0.034531
9	OnlineBackup	0.028112

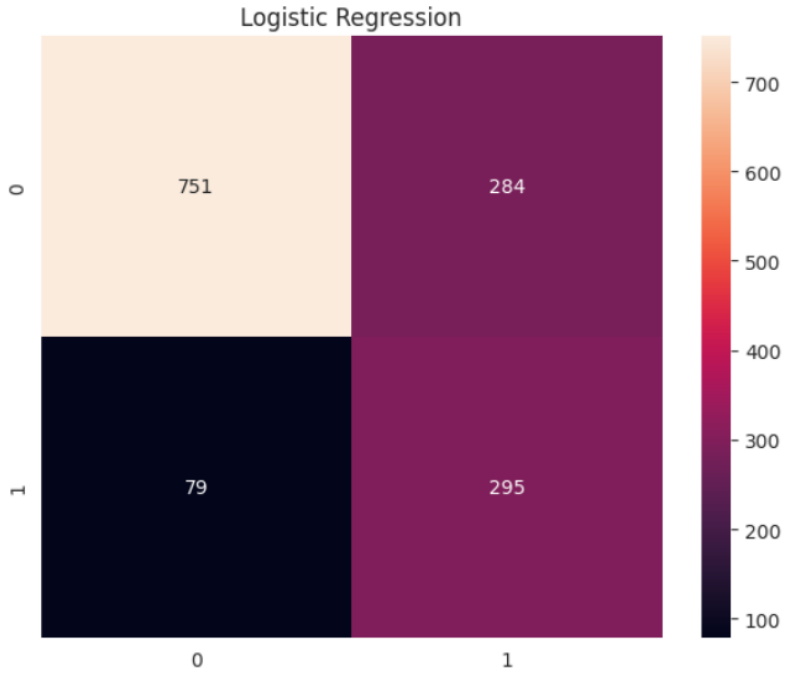

Top 10 Important Features
Contract
tenure
MonthlyCharges
TotalCharges
PaymentMethod
OnlineSecurity
TechSupport
InternetService
PaperlessBilling
OnlineBackup
Feature
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Importance


ROC Curve Comparison
Logistic Regression
Decision Tree
Random Forest
XGBoost
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0


XGBoost
867
168
149
225
0
1
0
1
800
700
600
500
400
300
200


Random Forest
866
169
157
217
0
1
0
1
800
700
600
500
400
300
200


Decision Tree
805
230
166
208
0
1
0
1
800
700
600
500
400
300
200


Logistic Regression
751
284
79
295
0
1
0
1
700
600
500
400
300
200
100

```
X = df.drop('Churn', axis=1)
y = df['Churn']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

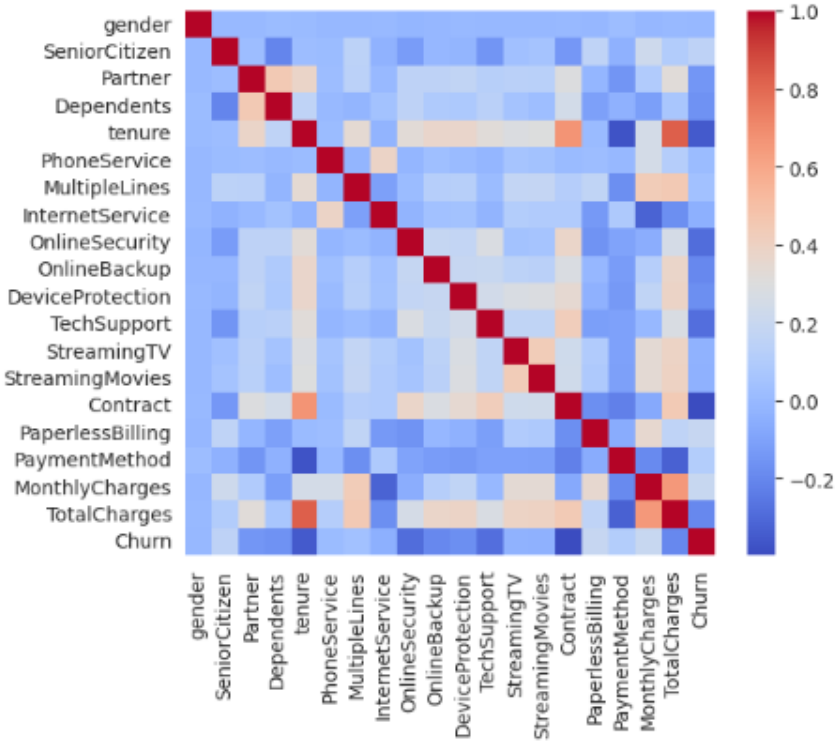
# scaling
scaler = StandardScaler()

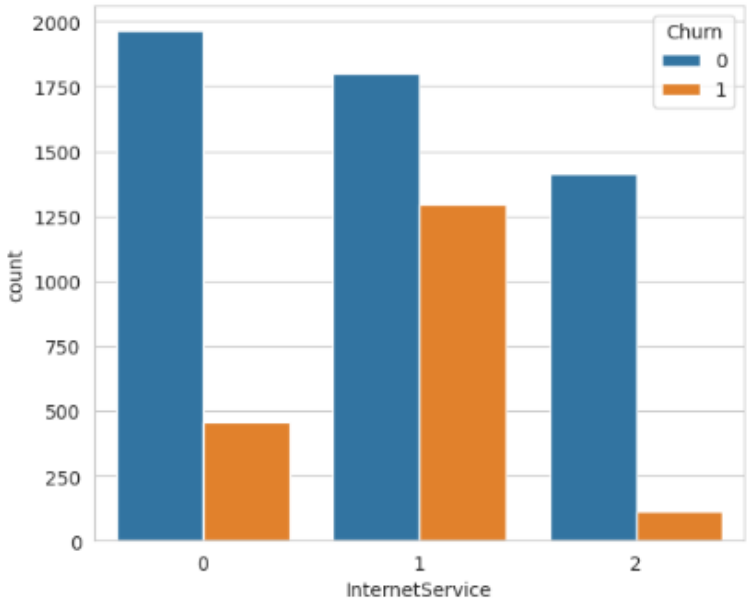
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

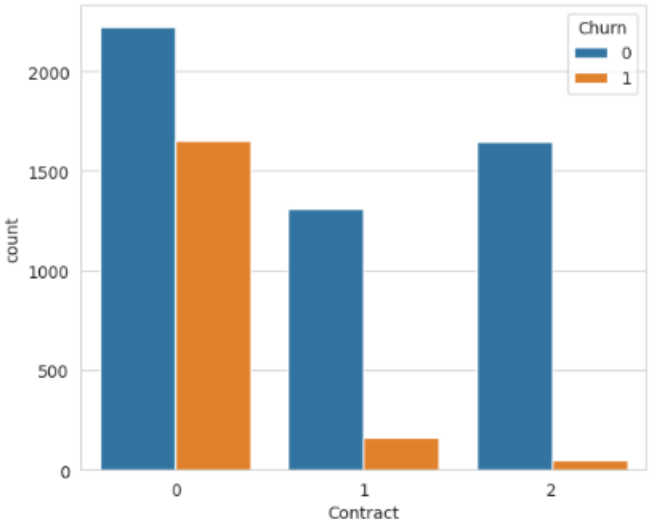
# handle imbalance
smote = SMOTE(random_state=42)

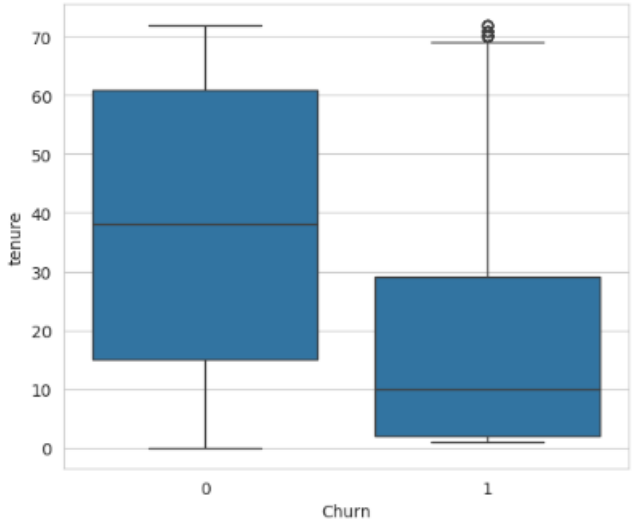
X_train, y_train = smote.fit_resample(
    X_train,
    y_train
)

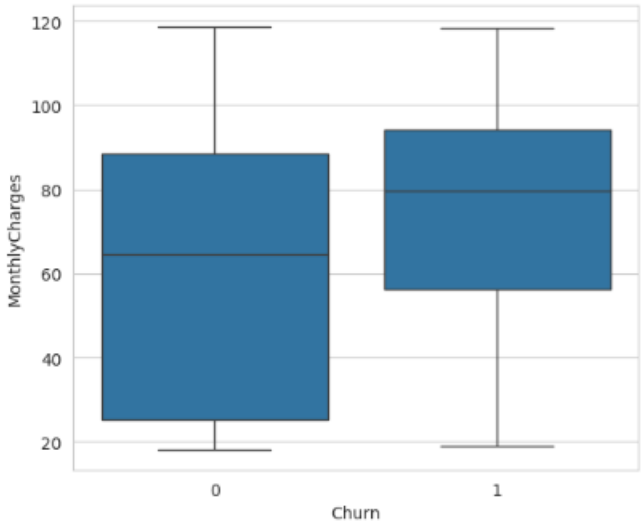
print(X_train.shape)
print(y_train.shape)
```

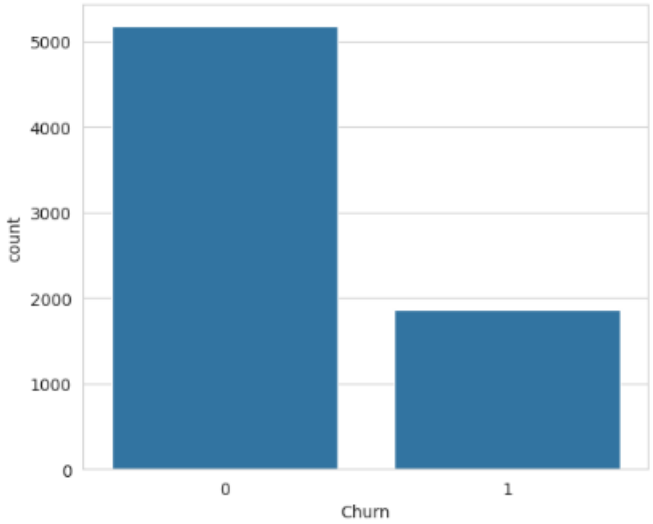



Churn
0
1
2000
1750
1500
1250
1000
750
500
250
0
count
0
1
2
InternetService


Churn
0
1
2000
1500
1000
500
0
count
0
1
2
Contract


70
60
50
40
30
20
10
0
tenure
0
1
Churn






```
df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')

print(df.head())

print("\nShape:")
print(df.shape)

print("\nData Types:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Shape:
(7043, 21)

Data Types:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
None
```

```
Missing Values:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```